Security

Concepts for keeping your cloud-native workload secure.

This section of the Kubernetes documentation aims to help you learn to run workloads more securely, and about the essential aspects of keeping a Kubernetes cluster secure.

Kubernetes is based on a cloud-native architecture, and draws on advice from the CNCF about good practice for cloud native information security.

Read Cloud Native Security and Kubernetes for the broader context about how to secure your cluster and the applications that you're running on it.

Kubernetes security mechanisms

Kubernetes includes several APIs and security controls, as well as ways to define policies that can form part of how you manage information security.

Control plane protection

A key security mechanism for any Kubernetes cluster is to control access to the Kubernetes API.

Kubernetes expects you to configure and use TLS to provide data encryption in transit within the control plane, and between the control plane and its clients. You can also enable encryption at rest for the data stored within Kubernetes control plane; this is separate from using encryption at rest for your own workloads' data, which might also be a good idea.

Secrets

The Secret API provides basic protection for configuration values that require confidentiality.

Workload protection

Enforce Pod security standards to ensure that Pods and their containers are isolated appropriately. You can also use RuntimeClasses to define custom isolation if you need it.

Network policies let you control network traffic between Pods, or between Pods and the network outside your cluster.

You can deploy security controls from the wider ecosystem to implement preventative or detective controls around Pods, their containers, and the images that run in them.

Admission control

Admission controllers are plugins that intercept Kubernetes API requests and can validate or mutate the requests based on specific fields in the request. Thoughtfully designing these controllers helps to avoid unintended disruptions as Kubernetes APIs change across version updates. For design considerations, see Admission Webhook Good Practices.

Auditing

Kubernetes audit logging provides a security-relevant, chronological set of records documenting the sequence of actions in a cluster. The cluster audits the activities generated by users, by applications that use the Kubernetes API, and by the control plane itself.

Cloud provider security

> **Note:** Items on this page refer to vendors external to Kubernetes. The Kubernetes project authors aren't responsible for those third-party products or projects. To add a

> vendor, product or project to this list, read the content guide before submitting a change. More information.

If you are running a Kubernetes cluster on your own hardware or a different cloud provider, consult your documentation for security best practices. Here are links to some of the popular cloud providers' security documentation:

IaaS Provider	Link
Alibaba Cloud	https://www.alibabacloud.com/trust-center
Amazon Web Services	https://aws.amazon.com/security
Google Cloud Platform	https://cloud.google.com/security
Huawei Cloud	https://www.huaweicloud.com/intl/en-us/securecenter/overallsafety
IBM Cloud	https://www.ibm.com/cloud/security
Microsoft Azure	https://docs.microsoft.com/en-us/azure/security/azure-security
Oracle Cloud Infrastructure	https://www.oracle.com/security
Tencent Cloud	https://www.tencentcloud.com/solutions/data-security-and-information-protection
VMware vSphere	https://www.vmware.com/solutions/security/hardening-guides

Policies

You can define security policies using Kubernetes-native mechanisms, such as NetworkPolicy (declarative control over network packet filtering) or ValidatingAdmissionPolicy (declarative restrictions on what changes someone can make using the Kubernetes API).

However, you can also rely on policy implementations from the wider ecosystem around Kubernetes. Kubernetes provides extension mechanisms to let those ecosystem projects implement their own policy controls on source code review, container image approval, API access controls, networking, and more.

For more information about policy mechanisms and Kubernetes, read Policies.

What's next

Learn about related Kubernetes security topics:

- Securing your cluster
- Known vulnerabilities in Kubernetes (and links to further information)
- Data encryption in transit for the control plane
- Data encryption at rest
- Controlling Access to the Kubernetes API
- Network policies for Pods
- Secrets in Kubernetes
- Pod security standards
- RuntimeClasses

Learn the context:

- Cloud Native Security and Kubernetes

Get certified:

- Certified Kubernetes Security Specialist certification and official training course.

Read more in this section:

1 - Cloud Native Security and Kubernetes

Concepts for keeping your cloud native workload secure.

Kubernetes is based on a cloud native architecture and draws on advice from the CNCF about good practices for cloud native information security.

Read on for an overview of how Kubernetes is designed to help you deploy a secure cloud native platform.

Cloud native information security

The CNCF white paper on cloud native security defines security controls and practices that are appropriate to different *lifecycle phases*.

Develop lifecycle phase

- Ensure the integrity of development environments.
- Design applications following good practices for information security, appropriate for your context.
- Consider end user security as part of solution design.

To achieve this, you can:

1. Adopt an architecture, such as zero trust, that minimizes attack surfaces, even for internal threats.
2. Define a code review process that considers security concerns.
3. Build a *threat model* of your system or application that identifies trust boundaries. Use that threat model to identify risks and determine how to treat them.
4. Incorporate advanced security automation, such as *fuzzing* and security chaos engineering, where it's justified.

Distribute lifecycle phase

- Ensure the security of the supply chain for container images you execute.

- Ensure the security of the supply chain for the cluster and other components that execute your application. For example, this might include an external database that your cloud native application uses for persistence.

To achieve this, you can:

1. Scan container images and other artifacts for known vulnerabilities.
2. Ensure that software distribution uses encryption in transit, with a chain of trust for the software source.
3. Adopt and follow processes to update dependencies when updates are available, especially in response to security announcements.
4. Use validation mechanisms such as digital certificates for supply chain assurance.
5. Subscribe to feeds and other mechanisms to alert you to security risks.
6. Restrict access to artifacts. Place container images in a private registry that only allows authorized clients to pull images.

Deploy lifecycle phase

Ensure appropriate restrictions on what can be deployed, who can deploy it, and where it can be deployed. You can enforce measures from the *distribute* phase, such as verifying the cryptographic identity of container image artifacts.

You can deploy different applications and cluster components into different namespaces. Containers and namespaces both provide isolation mechanisms that are relevant to information security.

When you deploy Kubernetes, you also set the foundation for your applications' runtime environment: a Kubernetes cluster (or multiple clusters). That infrastructure must provide the security guarantees that higher layers expect.

Runtime lifecycle phase

The Runtime phase comprises three critical areas: access, compute, and storage.

Runtime protection: access

The Kubernetes API is what makes your cluster work. Protecting this API is key to providing effective cluster security.

Other pages in the Kubernetes documentation have more detail about how to set up specific aspects of access control. The security checklist provides suggested basic checks for your cluster.

Beyond that, securing your cluster means implementing effective authentication and authorization for API access. Use ServiceAccounts to provide and manage security identities for workloads and cluster components.

Kubernetes uses TLS to protect API traffic; make sure to deploy the cluster using TLS (including for traffic between nodes and the control plane) and protect the encryption keys. If you use Kubernetes' own API for CertificateSigningRequests, pay special attention to restricting misuse there.

Runtime protection: compute

Containers provide two things: isolation between applications and a mechanism to combine those isolated applications to run on the same host computer. Those two aspects—isolation and aggregation—mean that runtime security involves identifying trade-offs and finding an appropriate balance.

Kubernetes relies on a container runtime to set up and run containers. The Kubernetes project does not recommend a specific container runtime, and you should make sure that the runtime(s) you choose meet your information security needs.

To protect your compute at runtime, you can:

1. Enforce Pod Security Standards for applications to help ensure they run with only the necessary privileges.

2. Run a specialized operating system on your nodes that is designed specifically for running containerized workloads. This is typically based on a read-only operating system (*immutable image*) that provides only the services essential for running containers.

   Container-specific operating systems help isolate system components and present a reduced attack surface in case of a container escape.

3. Define ResourceQuotas to fairly allocate shared resources, and use mechanisms such as LimitRanges to ensure that Pods specify their resource requirements.

4. Partition workloads across different nodes to improve isolation. Use node isolation mechanisms, either from Kubernetes itself or from the ecosystem, to ensure that Pods with different trust contexts run on separate sets of nodes.

5. Use a container runtime that provides security restrictions.

6. On Linux nodes, use a Linux security module such as AppArmor or seccomp.

Runtime protection: storage

To protect storage for your cluster and the applications that run there, you can:

1. Integrate your cluster with an external storage plugin that provides encryption at rest for volumes.
2. Enable encryption at rest for API objects.
3. Protect data durability using backups, and verify that you can restore them whenever needed.
4. Authenticate connections between cluster nodes and any network storage they rely upon.
5. Implement data encryption within your own application.

For encryption keys, generating these within specialized hardware provides the best protection against disclosure risks. A *hardware security module* can let you perform cryptographic operations without allowing the security key to be copied elsewhere.

Networking and security

You should also consider network security measures, such as NetworkPolicy or a service mesh. Some network plugins for Kubernetes provide encryption for your cluster network using technologies such as a virtual private network (VPN) overlay. By design, Kubernetes lets you use your own networking plugin for your cluster. If you use managed Kubernetes, the provider may have already selected a network plugin for you.

The network plugin you choose and the way you integrate it can have a strong impact on the security of information in transit.

Observability and runtime security

Kubernetes lets you extend your cluster with extra tooling. You can set up third party solutions to help you monitor or troubleshoot your applications and the clusters they are running. You also get some basic observability features built in to Kubernetes itself. Your code running in containers can generate logs, publish metrics, or provide other observability data; at deploy time, you need to make sure your cluster provides an appropriate level of protection there.

If you set up a metrics dashboard or something similar, review the chain of components that populate data into that dashboard, as well as the dashboard itself. Make sure that the whole chain is designed with enough resilience and integrity protection that you can rely on it even during an incident where your cluster might be degraded.

Where appropriate, deploy security measures below the Kubernetes layer, such as cryptographically measured boot or authenticated distribution of time (which helps ensure the fidelity of logs and audit records).

For a high-assurance environment, deploy cryptographic protections to ensure that logs are both tamper-proof and confidential.

What's next

Cloud native security

- CNCF white paper on cloud native security.
- CNCF white paper on good practices for securing a software supply chain.
- Fixing the Kubernetes clusterf**k: Understanding security from the kernel up (FOSDEM 2020)
- Kubernetes Security Best Practices (Kubernetes Forum Seoul 2019)
- Towards Measured Boot Out of the Box (Linux Security Summit 2016)

Kubernetes and information security

- Kubernetes security
- Securing your cluster
- Data encryption in transit for the control plane
- Data encryption at rest
- Secrets in Kubernetes
- Controlling Access to the Kubernetes API
- Network policies for Pods
- Pod security standards
- RuntimeClasses

2 - Pod Security Standards

A detailed look at the different policy levels defined in the Pod Security Standards.

The Pod Security Standards define three different *policies* to broadly cover the security spectrum. These policies are *cumulative* and range from highly-permissive to highly-restrictive. This guide outlines the requirements of each policy.

Profile	Description
Privileged	Unrestricted policy, providing the widest possible level of permissions. This policy allows for known privilege escalations.
Baseline	Minimally restrictive policy which prevents known privilege escalations. Allows the default (minimally specified) Pod configuration.
Restricted	Heavily restricted policy, following current Pod hardening best practices.

Profile Details

Privileged

The *Privileged* policy is purposely-open, and entirely unrestricted. This type of policy is typically aimed at system- and infrastructure-level workloads managed by privileged, trusted users.

The Privileged policy is defined by an absence of restrictions. If you define a Pod where the Privileged security policy applies, the Pod you define is able to bypass typical container isolation mechanisms. For example, you can define a Pod that has access to the node's host network.

Baseline

The *Baseline* policy is aimed at ease of adoption for common containerized workloads while preventing known privilege escalations. This policy is targeted at application operators and developers of non-critical applications. The following listed controls should be enforced/disallowed:

> **Note:**
> In this table, wildcards (`*`) indicate all elements in a list. For example, `spec.containers[*].securityContext` refers to the Security Context object for *all defined containers*. If any of the listed containers fails to meet the requirements, the entire pod will fail validation.

Control	Policy
HostProcess	Windows Pods offer the ability to run HostProcess containers whic access to the Windows host machine. Privileged access to the host Baseline policy. **FEATURE STATE:** Kubernetes v1.26 [stable] **Restricted Fields** • `spec.securityContext.windowsOptions.hostProcess` • `spec.containers[*].securityContext.windowsOptions.ho` • `spec.initContainers[*].securityContext.windowsOptior` • `spec.ephemeralContainers[*].securityContext.windowsC` **Allowed Values** • Undefined/nil • `false`

Host Namespaces	Sharing the host namespaces must be disallowed. **Restricted Fields** • `spec.hostNetwork` • `spec.hostPID` • `spec.hostIPC` **Allowed Values** • Undefined/nil • `false`
Privileged Containers	Privileged Pods disable most security mechanisms and must be di **Restricted Fields** • `spec.containers[*].securityContext.privileged` • `spec.initContainers[*].securityContext.privileged` • `spec.ephemeralContainers[*].securityContext.privileg` **Allowed Values** • Undefined/nil • `false`
Capabilities	Adding additional capabilities beyond those listed below must be d **Restricted Fields** • `spec.containers[*].securityContext.capabilities.add` • `spec.initContainers[*].securityContext.capabilities.` • `spec.ephemeralContainers[*].securityContext.capabili` **Allowed Values** • Undefined/nil • `AUDIT_WRITE` • `CHOWN` • `DAC_OVERRIDE` • `FOWNER`

	• FSETID • KILL • MKNOD • NET_BIND_SERVICE • SETFCAP • SETGID • SETPCAP • SETUID • SYS_CHROOT
HostPath Volumes	HostPath volumes must be forbidden. **Restricted Fields** • spec.volumes[*].hostPath **Allowed Values** • Undefined/nil
Host Ports	HostPorts should be disallowed entirely (recommended) or restric **Restricted Fields** • spec.containers[*].ports[*].hostPort • spec.initContainers[*].ports[*].hostPort • spec.ephemeralContainers[*].ports[*].hostPort **Allowed Values** • Undefined/nil • Known list (not supported by the built-in Pod Security Admissi • 0
Host Probes / Lifecycle Hooks (v1.34+)	The Host field in probes and lifecycle hooks must be disallowed. **Restricted Fields** • spec.containers[*].livenessProbe.httpGet.host • spec.containers[*].readinessProbe.httpGet.host

- `spec.containers[*].startupProbe.httpGet.host`
- `spec.containers[*].livenessProbe.tcpSocket.host`
- `spec.containers[*].readinessProbe.tcpSocket.host`
- `spec.containers[*].startupProbe.tcpSocket.host`
- `spec.containers[*].lifecycle.postStart.tcpSocket.hos`
- `spec.containers[*].lifecycle.preStop.tcpSocket.host`
- `spec.containers[*].lifecycle.postStart.httpGet.host`
- `spec.containers[*].lifecycle.preStop.httpGet.host`
- `spec.initContainers[*].livenessProbe.httpGet.host`
- `spec.initContainers[*].readinessProbe.httpGet.host`
- `spec.initContainers[*].startupProbe.httpGet.host`
- `spec.initContainers[*].livenessProbe.tcpSocket.host`
- `spec.initContainers[*].readinessProbe.tcpSocket.host`
- `spec.initContainers[*].startupProbe.tcpSocket.host`
- `spec.initContainers[*].lifecycle.postStart.tcpSocket`
- `spec.initContainers[*].lifecycle.preStop.tcpSocket.h`
- `spec.initContainers[*].lifecycle.postStart.httpGet.h`
- `spec.initContainers[*].lifecycle.preStop.httpGet.hos`

Allowed Values

- Undefined/nil
- ""

AppArmor	On supported hosts, the `RuntimeDefault` AppArmor profile is ap baseline policy should prevent overriding or disabling the default A restrict overrides to an allowed set of profiles. **Restricted Fields** • `spec.securityContext.appArmorProfile.type` • `spec.containers[*].securityContext.appArmorProfile.t` • `spec.initContainers[*].securityContext.appArmorProfi` • `spec.ephemeralContainers[*].securityContext.appArmor` **Allowed Values** • Undefined/nil

- `RuntimeDefault`
- `Localhost`

- `metadata.annotations["container.apparmor.security.be`

Allowed Values

- Undefined/nil
- `runtime/default`
- `localhost/*`

SELinux

Setting the SELinux type is restricted, and setting a custom SELinux forbidden.

Restricted Fields

- `spec.securityContext.seLinuxOptions.type`
- `spec.containers[*].securityContext.seLinuxOptions.ty`
- `spec.initContainers[*].securityContext.seLinuxOptior`
- `spec.ephemeralContainers[*].securityContext.seLinuxC`

Allowed Values

- Undefined/""
- `container_t`
- `container_init_t`
- `container_kvm_t`
- `container_engine_t` (since Kubernetes 1.31)

Restricted Fields

- `spec.securityContext.seLinuxOptions.user`
- `spec.containers[*].securityContext.seLinuxOptions.us`
- `spec.initContainers[*].securityContext.seLinuxOptior`
- `spec.ephemeralContainers[*].securityContext.seLinuxC`
- `spec.securityContext.seLinuxOptions.role`
- `spec.containers[*].securityContext.seLinuxOptions.rc`
- `spec.initContainers[*].securityContext.seLinuxOptior`

- `spec.ephemeralContainers[*].securityContext.seLinuxO`

Allowed Values

- Undefined/""

`/proc` Mount Type	The default `/proc` masks are set up to reduce attack surface, and **Restricted Fields** • `spec.containers[*].securityContext.procMount` • `spec.initContainers[*].securityContext.procMount` • `spec.ephemeralContainers[*].securityContext.procMoun  **Allowed Values**  • Undefined/nil • `Default`
Seccomp	Seccomp profile must not be explicitly set to `Unconfined`. **Restricted Fields** • `spec.securityContext.seccompProfile.type` • `spec.containers[*].securityContext.seccompProfile.ty` • `spec.initContainers[*].securityContext.seccompProfil` • `spec.ephemeralContainers[*].securityContext.seccompP` **Allowed Values** • Undefined/nil • `RuntimeDefault` • `Localhost`
Sysctls	Sysctls can disable security mechanisms or affect all containers on disallowed except for an allowed "safe" subset. A sysctl is consider namespaced in the container or the Pod, and it is isolated from otl the same Node. **Restricted Fields**

- `spec.securityContext.sysctls[*].name`

Allowed Values

- Undefined/nil
- `kernel.shm_rmid_forced`
- `net.ipv4.ip_local_port_range`
- `net.ipv4.ip_unprivileged_port_start`
- `net.ipv4.tcp_syncookies`
- `net.ipv4.ping_group_range`
- `net.ipv4.ip_local_reserved_ports` (since Kubernetes 1.2
- `net.ipv4.tcp_keepalive_time` (since Kubernetes 1.29)
- `net.ipv4.tcp_fin_timeout` (since Kubernetes 1.29)
- `net.ipv4.tcp_keepalive_intvl` (since Kubernetes 1.29)
- `net.ipv4.tcp_keepalive_probes` (since Kubernetes 1.29)

Restricted

The *Restricted* policy is aimed at enforcing current Pod hardening best practices, at the expense of some compatibility. It is targeted at operators and developers of security-critical applications, as well as lower-trust users. The following listed controls should be enforced/disallowed:

> **Note:**
> In this table, wildcards (`*`) indicate all elements in a list. For example, `spec.containers[*].securityContext` refers to the Security Context object for *all defined containers*. If any of the listed containers fails to meet the requirements, the entire pod will fail validation.

Control	Policy
Everything from the Baseline policy	
Volume Types	The Restricted policy only permits the following volun **Restricted Fields**

	• `spec.volumes[*]` **Allowed Values** Every item in the `spec.volumes[*]` list must set one null value: • `spec.volumes[*].configMap` • `spec.volumes[*].csi` • `spec.volumes[*].downwardAPI` • `spec.volumes[*].emptyDir` • `spec.volumes[*].ephemeral` • `spec.volumes[*].persistentVolumeClaim` • `spec.volumes[*].projected` • `spec.volumes[*].secret`
Privilege Escalation (v1.8+)	Privilege escalation (such as via set-user-ID or set-gro allowed. *This is Linux only policy in v1.25+ (`spec.os.n`* **Restricted Fields** • `spec.containers[*].securityContext.allowP` • `spec.initContainers[*].securityContext.al` • `spec.ephemeralContainers[*].securityConte` **Allowed Values** • `false`
Running as Non-root	Containers must be required to run as non-root users **Restricted Fields** • `spec.securityContext.runAsNonRoot` • `spec.containers[*].securityContext.runAsN` • `spec.initContainers[*].securityContext.ru` • `spec.ephemeralContainers[*].securityConte` **Allowed Values**

	• `true` The container fields may be undefined/ `nil` if the pod-level `spec.securityContext.runAsNonRoot` is set to `true`.
Running as Non-root user (v1.23+)	Containers must not set `runAsUser` to 0 **Restricted Fields** • `spec.securityContext.runAsUser` • `spec.containers[*].securityContext.runAsU` • `spec.initContainers[*].securityContext.ru` • `spec.ephemeralContainers[*].securityConte` **Allowed Values** • any non-zero value • `undefined/null`
Seccomp (v1.19+)	Seccomp profile must be explicitly set to one of the a `Unconfined` profile and the *absence* of a profile are *policy in v1.25+ (`spec.os.name != windows`)* **Restricted Fields** • `spec.securityContext.seccompProfile.type` • `spec.containers[*].securityContext.seccom` • `spec.initContainers[*].securityContext.se` • `spec.ephemeralContainers[*].securityConte` **Allowed Values** • `RuntimeDefault` • `Localhost` The container fields may be undefined/ `nil` if the pod-level `spec.securityContext.seccompProfile.type` field is set a level field may be undefined/ `nil` if _all_ container- level fiel
Capabilities (v1.22+)	Containers must drop `ALL` capabilities, and are only `NET_BIND_SERVICE` capability. *This is Linux only polic `"windows"`)*

Restricted Fields

- `spec.containers[*].securityContext.capabi`
- `spec.initContainers[*].securityContext.ca`
- `spec.ephemeralContainers[*].securityConte`

Allowed Values

- Any list of capabilities that includes `ALL`

Restricted Fields

- `spec.containers[*].securityContext.capabi`
- `spec.initContainers[*].securityContext.ca`
- `spec.ephemeralContainers[*].securityConte`

Allowed Values

- Undefined/nil
- `NET_BIND_SERVICE`

Policy Instantiation

Decoupling policy definition from policy instantiation allows for a common understanding and consistent language of policies across clusters, independent of the underlying enforcement mechanism.

As mechanisms mature, they will be defined below on a per-policy basis. The methods of enforcement of individual policies are not defined here.

Pod Security Admission Controller

- Privileged namespace
- Baseline namespace
- Restricted namespace

Alternatives

> **Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the content guide before submitting a change. More information.

Other alternatives for enforcing policies are being developed in the Kubernetes ecosystem, such as:

- Kubewarden
- Kyverno
- OPA Gatekeeper

Pod OS field

Kubernetes lets you use nodes that run either Linux or Windows. You can mix both kinds of node in one cluster. Windows in Kubernetes has some limitations and differentiators from Linux-based workloads. Specifically, many of the Pod `securityContext` fields have no effect on Windows.

> **Note:**
> Kubelets prior to v1.24 don't enforce the pod OS field, and if a cluster has nodes on versions earlier than v1.24 the Restricted policies should be pinned to a version prior to v1.25.

Restricted Pod Security Standard changes

Another important change, made in Kubernetes v1.25 is that the *Restricted* policy has been updated to use the `pod.spec.os.name` field. Based on the OS name, certain policies that are specific to a particular OS can be relaxed for the other OS.

OS-specific policy controls

Restrictions on the following controls are only required if `.spec.os.name` is not `windows`:

- Privilege Escalation
- Seccomp

- Linux Capabilities

User namespaces

User Namespaces are a Linux-only feature to run workloads with increased isolation. How they work together with Pod Security Standards is described in the documentation for Pods that use user namespaces.

FAQ

Why isn't there a profile between Privileged and Baseline?

The three profiles defined here have a clear linear progression from most secure (Restricted) to least secure (Privileged), and cover a broad set of workloads. Privileges required above the Baseline policy are typically very application specific, so we do not offer a standard profile in this niche. This is not to say that the privileged profile should always be used in this case, but that policies in this space need to be defined on a case-by-case basis.

SIG Auth may reconsider this position in the future, should a clear need for other profiles arise.

What's the difference between a security profile and a security context?

Security Contexts configure Pods and Containers at runtime. Security contexts are defined as part of the Pod and container specifications in the Pod manifest, and represent parameters to the container runtime.

Security profiles are control plane mechanisms to enforce specific settings in the Security Context, as well as other related parameters outside the Security Context. As of July 2021, Pod Security Policies are deprecated in favor of the built-in Pod Security Admission Controller.

What about sandboxed Pods?

There is currently no API standard that controls whether a Pod is considered sandboxed or not. Sandbox Pods may be identified by the use of a sandboxed runtime (such as gVisor or Kata Containers), but there is no standard definition of what a sandboxed runtime is.

The protections necessary for sandboxed workloads can differ from others. For example, the need to restrict privileged permissions is lessened when the workload is isolated from the underlying kernel. This allows for workloads requiring heightened permissions to still be isolated.

Additionally, the protection of sandboxed workloads is highly dependent on the method of sandboxing. As such, no single recommended profile is recommended for all sandboxed workloads.

3 - Pod Security Admission

An overview of the Pod Security Admission Controller, which can enforce the Pod Security Standards.

FEATURE STATE: Kubernetes v1.25 [stable]

The Kubernetes Pod Security Standards define different isolation levels for Pods. These standards let you define how you want to restrict the behavior of pods in a clear, consistent fashion.

Kubernetes offers a built-in *Pod Security* admission controller to enforce the Pod Security Standards. Pod security restrictions are applied at the namespace level when pods are created.

Built-in Pod Security admission enforcement

This page is part of the documentation for Kubernetes v1.36. If you are running a different version of Kubernetes, consult the documentation for that release.

Pod Security levels

Pod Security admission places requirements on a Pod's Security Context and other related fields according to the three levels defined by the Pod Security Standards: `privileged`, `baseline`, and `restricted`. Refer to the Pod Security Standards page for an in-depth look at those requirements.

Pod Security Admission labels for namespaces

Once the feature is enabled or the webhook is installed, you can configure namespaces to define the admission control mode you want to use for pod security in each namespace. Kubernetes defines a set of labels that you can set to define which of the predefined Pod Security Standard levels you want to use for a namespace. The label you select defines what action the control plane takes if a potential violation is detected:

Mode	Description
enforce	Policy violations will cause the pod to be rejected.
audit	Policy violations will trigger the addition of an audit annotation to the event recorded in the audit log, but are otherwise allowed.
warn	Policy violations will trigger a user-facing warning, but are otherwise allowed.

A namespace can configure any or all modes, or even set a different level for different modes.

For each mode, there are two labels that determine the policy used:

```
# The per-mode level label indicates which policy level to apply for the mode.
#
# MODE must be one of `enforce`, `audit`, or `warn`.
# LEVEL must be one of `privileged`, `baseline`, or `restricted`.
pod-security.kubernetes.io/<MODE>: <LEVEL>

# Optional: per-mode version label that can be used to pin the policy to the
# version that shipped with a given Kubernetes minor version (for example v1.36).
#
# MODE must be one of `enforce`, `audit`, or `warn`.
# VERSION must be a valid Kubernetes minor version, or `latest`.
pod-security.kubernetes.io/<MODE>-version: <VERSION>
```

Check out Enforce Pod Security Standards with Namespace Labels to see example usage.

Workload resources and Pod templates

Pods are often created indirectly, by creating a workload object such as a Deployment or Job. The workload object defines a *Pod template* and a controller for the workload resource creates Pods based on that template. To help catch violations early, both the audit and warning modes are applied to the workload resources. However, enforce mode is **not** applied to workload resources, only to the resulting pod objects.

Exemptions

You can define *exemptions* from pod security enforcement in order to allow the creation of pods that would have otherwise been prohibited due to the policy associated with a given namespace. Exemptions can be statically configured in the Admission Controller configuration.

Exemptions must be explicitly enumerated. Requests meeting exemption criteria are *ignored* by the Admission Controller (all `enforce`, `audit` and `warn` behaviors are skipped). Exemption dimensions include:

- **Usernames:** requests from users with an exempt authenticated (or impersonated) username are ignored.
- **RuntimeClassNames:** pods and workload resources specifying an exempt runtime class name are ignored.
- **Namespaces:** pods and workload resources in an exempt namespace are ignored.

> **Caution:**
>
> Most pods are created by a controller in response to a workload resource, meaning that exempting an end user will only exempt them from enforcement when creating pods directly, but not when creating a workload resource. Controller service accounts (such as `system:serviceaccount:kube-system:replicaset-controller`) should generally not be exempted, as doing so would implicitly exempt any user that can create the corresponding workload resource.

Updates to the following pod fields are exempt from policy checks, meaning that if a pod update request only changes these fields, it will not be denied even if the pod is in violation of the current policy level:

- Any metadata updates **except** changes to the seccomp or AppArmor annotations:
  - `seccomp.security.alpha.kubernetes.io/pod` (deprecated)
  - `container.seccomp.security.alpha.kubernetes.io/*` (deprecated)
  - `container.apparmor.security.beta.kubernetes.io/*` (deprecated)
- Valid updates to `.spec.activeDeadlineSeconds`
- Valid updates to `.spec.tolerations`

Metrics

Here are the Prometheus metrics exposed by kube-apiserver:

- `pod_security_errors_total` : This metric indicates the number of errors preventing normal evaluation. Non-fatal errors may result in the latest restricted profile being used for enforcement.
- `pod_security_evaluations_total` : This metric indicates the number of policy evaluations that have occurred, not counting ignored or exempt requests during exporting.
- `pod_security_exemptions_total` : This metric indicates the number of exempt requests, not counting ignored or out of scope requests.

What's next

- Pod Security Standards
- Enforcing Pod Security Standards
- Enforce Pod Security Standards by Configuring the Built-in Admission Controller
- Enforce Pod Security Standards with Namespace Labels

If you are running an older version of Kubernetes and want to upgrade to a version of Kubernetes that does not include PodSecurityPolicies, read migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller.

4 - Service Accounts

Learn about ServiceAccount objects in Kubernetes.

This page introduces the ServiceAccount object in Kubernetes, providing information about how service accounts work, use cases, limitations, alternatives, and links to resources for additional guidance.

What are service accounts?

A service account is a type of non-human account that, in Kubernetes, provides a distinct identity in a Kubernetes cluster. Application Pods, system components, and entities inside and outside the cluster can use a specific ServiceAccount's credentials to identify as that ServiceAccount. This identity is useful in various situations, including authenticating to the API server or implementing identity-based security policies.

Service accounts exist as ServiceAccount objects in the API server. Service accounts have the following properties:

- **Namespaced:** Each service account is bound to a Kubernetes namespace. Every namespace gets a `default` ServiceAccount upon creation.
- **Lightweight:** Service accounts exist in the cluster and are defined in the Kubernetes API. You can quickly create service accounts to enable specific tasks.
- **Portable:** A configuration bundle for a complex containerized workload might include service account definitions for the system's components. The lightweight nature of service accounts and the namespaced identities make the configurations portable.

Service accounts are different from user accounts, which are authenticated human users in the cluster. By default, user accounts don't exist in the Kubernetes API server; instead, the API server treats user identities as opaque data. You can authenticate as a user account using multiple methods. Some Kubernetes distributions might add custom extension APIs to represent user accounts in the API server.

Description	ServiceAccount	User or group
Location	Kubernetes API (ServiceAccount object)	External

Description	ServiceAccount	User or group
Access control	Kubernetes RBAC or other authorization mechanisms	Kubernetes RBAC or other identity and access management mechanisms
Intended use	Workloads, automation	People

Default service accounts

When you create a cluster, Kubernetes automatically creates a ServiceAccount object named `default` for every namespace in your cluster. The `default` service accounts in each namespace get no permissions by default other than the default API discovery permissions that Kubernetes grants to all authenticated principals if role-based access control (RBAC) is enabled. If you delete the `default` ServiceAccount object in a namespace, the control plane replaces it with a new one.

If you deploy a Pod in a namespace, and you don't manually assign a ServiceAccount to the Pod, Kubernetes assigns the `default` ServiceAccount for that namespace to the Pod.

Use cases for Kubernetes service accounts

As a general guideline, you can use service accounts to provide identities in the following scenarios:

- Your Pods need to communicate with the Kubernetes API server, for example in situations such as the following:
  - Providing read-only access to sensitive information stored in Secrets.
  - Granting cross-namespace access, such as allowing a Pod in namespace `example` to read, list, and watch for Lease objects in the `kube-node-lease` namespace.
- Your Pods need to communicate with an external service. For example, a workload Pod requires an identity for a commercially available cloud API, and the commercial provider allows configuring a suitable trust relationship.
- Authenticating to a private image registry using an `imagePullSecret`.
- An external service needs to communicate with the Kubernetes API server. For example, authenticating to the cluster as part of a CI/CD pipeline.
- You use third-party security software in your cluster that relies on the ServiceAccount identity of different Pods to group those Pods into different contexts.

How to use service accounts

To use a Kubernetes service account, you do the following:

1. Create a ServiceAccount object using a Kubernetes client like `kubectl` or a manifest that defines the object.

2. Grant permissions to the ServiceAccount object using an authorization mechanism such as RBAC.

3. Assign the ServiceAccount object to Pods during Pod creation.

   If you're using the identity from an external service, retrieve the ServiceAccount token and use it from that service instead.

For instructions, refer to Configure Service Accounts for Pods.

Grant permissions to a ServiceAccount

You can use the built-in Kubernetes role-based access control (RBAC) mechanism to grant the minimum permissions required by each service account. You create a *role*, which grants access, and then *bind* the role to your ServiceAccount. RBAC lets you define a minimum set of permissions so that the service account permissions follow the principle of least privilege. Pods that use that service account don't get more permissions than are required to function correctly.

For instructions, refer to ServiceAccount permissions.

Cross-namespace access using a ServiceAccount

You can use RBAC to allow service accounts in one namespace to perform actions on resources in a different namespace in the cluster. For example, consider a scenario where you have a service account and Pod in the `dev` namespace and you want your Pod to see Jobs running in the `maintenance` namespace. You could create a Role object that grants permissions to list Job objects. Then, you'd create a RoleBinding object in the `maintenance` namespace to bind the Role to the ServiceAccount object. Now, Pods in the `dev` namespace can list Job objects in the `maintenance` namespace using that service account.

Assign a ServiceAccount to a Pod

To assign a ServiceAccount to a Pod, you set the `spec.serviceAccountName` field in the Pod specification. Kubernetes then automatically provides the credentials for that

ServiceAccount to the Pod. In v1.22 and later, Kubernetes gets a short-lived, **automatically rotating** token using the `TokenRequest` API and mounts the token as a projected volume.

By default, Kubernetes provides the Pod with the credentials for an assigned ServiceAccount, whether that is the `default` ServiceAccount or a custom ServiceAccount that you specify.

To prevent Kubernetes from automatically injecting credentials for a specified ServiceAccount or the `default` ServiceAccount, set the `automountServiceAccountToken` field in your Pod specification to `false`.

In versions earlier than 1.22, Kubernetes provides a long-lived, static token to the Pod as a Secret.

Manually retrieve ServiceAccount credentials

If you need the credentials for a ServiceAccount to mount in a non-standard location, or for an audience that isn't the API server, use one of the following methods:

- TokenRequest API (recommended): Request a short-lived service account token from within your own *application code*. The token expires automatically and can rotate upon expiration. If you have a legacy application that is not aware of Kubernetes, you could use a sidecar container within the same pod to fetch these tokens and make them available to the application workload.
- Token Volume Projection (also recommended): In Kubernetes v1.20 and later, use the Pod specification to tell the kubelet to add the service account token to the Pod as a *projected volume*. Projected tokens expire automatically, and the kubelet rotates the token before it expires.
- Service Account Token Secrets (not recommended): You can mount service account tokens as Kubernetes Secrets in Pods. These tokens don't expire and don't rotate. In versions prior to v1.24, a permanent token was automatically created for each service account. This method is not recommended anymore, especially at scale, because of the risks associated with static, long-lived credentials. The LegacyServiceAccountTokenNoAutoGeneration feature gate (which was enabled by default from Kubernetes v1.24 to v1.26), prevented Kubernetes from automatically creating these tokens for ServiceAccounts. The feature gate is removed in v1.27, because it was elevated to GA status; you can still create indefinite service account tokens manually, but should take into account the security implications.

Node audience restriction for service account tokens

ⓘ **FEATURE STATE:** Kubernetes v1.33 [beta](enabled by default)

When the `ServiceAccountNodeAudienceRestriction` feature gate is enabled, the NodeRestriction admission plugin limits which audiences a kubelet can request when creating service account tokens via the `TokenRequest` API. By default, the kubelet can only request tokens for audiences already referenced by pods on that node (through projected service account token volumes or CSI driver token requests). Administrators can grant kubelets access to additional audiences using RBAC rules with the `request-serviceaccounts-token-audience` verb.

This restriction applies only to kubelets (node identities) and does not affect other callers of the `TokenRequest` API. For details and RBAC examples, see Service account token audience restriction.

> **Note:**
> For applications running outside your Kubernetes cluster, you might be considering creating a long-lived ServiceAccount token that is stored in a Secret. This allows authentication, but the Kubernetes project recommends you avoid this approach. Long-lived bearer tokens represent a security risk as, once disclosed, the token can be misused. Instead, consider using an alternative. For example, your external application can authenticate using a well-protected private key `and` a certificate, or using a custom mechanism such as an authentication webhook that you implement yourself.
>
> You can also use TokenRequest to obtain short-lived tokens for your external application.

Restricting access to Secrets (deprecated)

ⓘ **FEATURE STATE:** Kubernetes v1.32 [deprecated]

> **Note:**

> `kubernetes.io/enforce-mountable-secrets` is deprecated since Kubernetes v1.32. Use separate namespaces to isolate access to mounted secrets.

Kubernetes provides an annotation called `kubernetes.io/enforce-mountable-secrets` that you can add to your ServiceAccounts. When this annotation is applied, the ServiceAccount's secrets can only be mounted on specified types of resources, enhancing the security posture of your cluster.

You can add the annotation to a ServiceAccount using a manifest:

```yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    kubernetes.io/enforce-mountable-secrets: "true"
  name: my-serviceaccount
  namespace: my-namespace
```

When this annotation is set to "true", the Kubernetes control plane ensures that the Secrets from this ServiceAccount are subject to certain mounting restrictions.

1. The name of each Secret that is mounted as a volume in a Pod must appear in the `secrets` field of the Pod's ServiceAccount.
2. The name of each Secret referenced using `envFrom` in a Pod must also appear in the `secrets` field of the Pod's ServiceAccount.
3. The name of each Secret referenced using `imagePullSecrets` in a Pod must also appear in the `secrets` field of the Pod's ServiceAccount.

By understanding and enforcing these restrictions, cluster administrators can maintain a tighter security profile and ensure that secrets are accessed only by the appropriate resources.

Authenticating service account credentials

ServiceAccounts use signed JSON Web Tokens (JWTs) to authenticate to the Kubernetes API server, and to any other system where a trust relationship exists. Depending on how the token was issued (either time-limited using a `TokenRequest` or using a legacy

mechanism with a Secret), a ServiceAccount token might also have an expiry time, an audience, and a time after which the token *starts* being valid. When a client that is acting as a ServiceAccount tries to communicate with the Kubernetes API server, the client includes an `Authorization: Bearer <token>` header with the HTTP request. The API server checks the validity of that bearer token as follows:

1. Checks the token signature.
2. Checks whether the token has expired.
3. Checks whether object references in the token claims are currently valid.
4. Checks whether the token is currently valid.
5. Checks the audience claims.

The TokenRequest API produces *bound tokens* for a ServiceAccount. This binding is linked to the lifetime of the client, such as a Pod, that is acting as that ServiceAccount. See Token Volume Projection for an example of a bound pod service account token's JWT schema and payload.

For tokens issued using the `TokenRequest` API, the API server also checks that the specific object reference that is using the ServiceAccount still exists, matching by the unique ID of that object. For legacy tokens that are mounted as Secrets in Pods, the API server checks the token against the Secret.

For more information about the authentication process, refer to Authentication.

Authenticating service account credentials in your own code

If you have services of your own that need to validate Kubernetes service account credentials, you can use the following methods:

- TokenReview API (recommended)
- OIDC discovery

The Kubernetes project recommends that you use the TokenReview API, because this method invalidates tokens that are bound to API objects such as Secrets, ServiceAccounts, Pods or Nodes when those objects are deleted. For example, if you delete the Pod that contains a projected ServiceAccount token, the cluster invalidates that token immediately and a TokenReview immediately fails. If you use OIDC validation instead, your clients continue to treat the token as valid until the token reaches its expiration timestamp.

Your application should always define the audience that it accepts, and should check that the token's audiences match the audiences that the application expects. This helps to

minimize the scope of the token so that it can only be used in your application and nowhere else.

Alternatives

- Issue your own tokens using another mechanism, and then use Webhook Token Authentication to validate bearer tokens using your own validation service.
- Provide your own identities to Pods.
  - Use the SPIFFE CSI driver plugin to provide SPIFFE SVIDs as X.509 certificate pairs to Pods.

    > ⃠ This item links to a third party project or product that is not part of Kubernetes itself. More information

  - Use a service mesh such as Istio to provide certificates to Pods.
- Authenticate from outside the cluster to the API server without using service account tokens:
  - Configure the API server to accept OpenID Connect (OIDC) tokens from your identity provider.
  - Use service accounts or user accounts created using an external Identity and Access Management (IAM) service, such as from a cloud provider, to authenticate to your cluster.
  - Use the CertificateSigningRequest API with client certificates.
- Configure the kubelet to retrieve credentials from an image registry.
- Use a Device Plugin to access a virtual Trusted Platform Module (TPM), which then allows authentication using a private key.

What's next

- Learn how to manage your ServiceAccounts as a cluster administrator.
- Learn how to assign a ServiceAccount to a Pod.
- Read the ServiceAccount API reference.

5 - Pod Security Policies

> **Removed feature**
>
> PodSecurityPolicy was deprecated in Kubernetes v1.21, and removed from Kubernetes in v1.25.

Instead of using PodSecurityPolicy, you can enforce similar restrictions on Pods using either or both:

- Pod Security Admission
- a 3rd party admission plugin, that you deploy and configure yourself

For a migration guide, see Migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller. For more information on the removal of this API, see PodSecurityPolicy Deprecation: Past, Present, and Future.

If you are not running Kubernetes v1.36, check the documentation for your version of Kubernetes.

6 - Security For Linux Nodes

This page describes security considerations and best practices specific to the Linux operating system.

Protection for Secret data on nodes

On Linux nodes, memory-backed volumes (such as `secret` volume mounts, or `emptyDir` with `medium: Memory`) are implemented with a `tmpfs` filesystem.

If you have swap configured and use an older Linux kernel (or a current kernel and an unsupported configuration of Kubernetes), **memory** backed volumes can have data written to persistent storage.

The Linux kernel officially supports the `noswap` option from version 6.3, therefore it is recommended the used kernel version is 6.3 or later, or supports the `noswap` option via a backport, if swap is enabled on the node.

Read swap memory management for more info.

7 - Security For Windows Nodes

This page describes security considerations and best practices specific to the Windows operating system.

Protection for Secret data on nodes

On Windows, data from Secrets are written out in clear text onto the node's local storage (as compared to using tmpfs / in-memory filesystems on Linux). As a cluster operator, you should take both of the following additional measures:

1. Use file ACLs to secure the Secrets' file location.
2. Apply volume-level encryption using BitLocker.

Container users

RunAsUsername can be specified for Windows Pods or containers to execute the container processes as specific user. This is roughly equivalent to RunAsUser.

Windows containers offer two default user accounts, ContainerUser and ContainerAdministrator. The differences between these two user accounts are covered in When to use ContainerAdmin and ContainerUser user accounts within Microsoft's *Secure Windows containers* documentation.

Local users can be added to container images during the container build process.

> **Note:**
> - Nano Server based images run as `ContainerUser` by default
> - Server Core based images run as `ContainerAdministrator` by default

Windows containers can also run as Active Directory identities by utilizing Group Managed Service Accounts

Pod-level security isolation

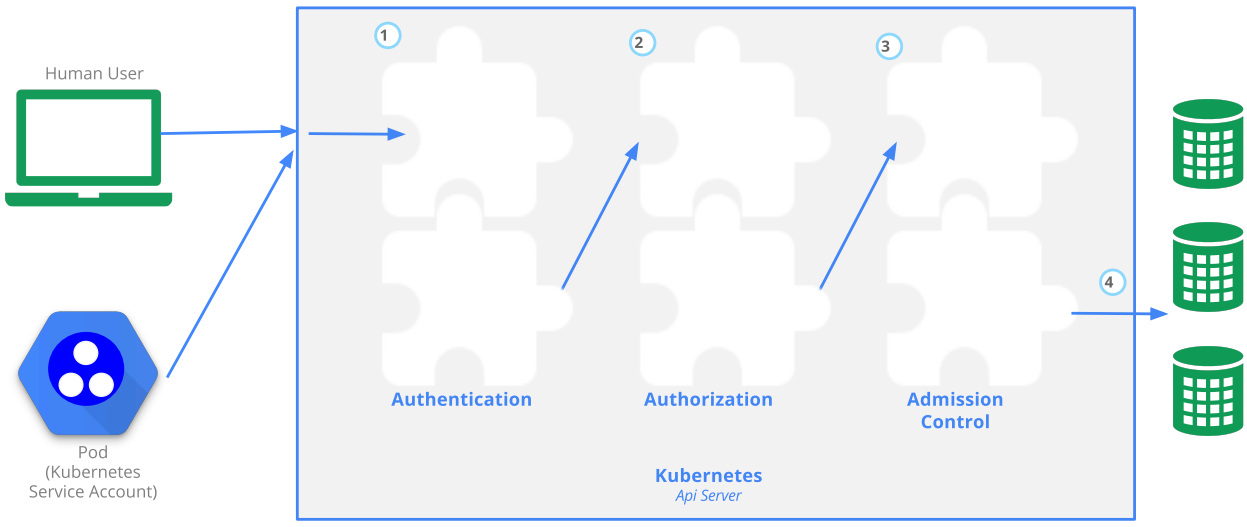
Linux-specific pod security context mechanisms (such as SELinux, AppArmor, Seccomp, or custom POSIX capabilities) are not supported on Windows nodes.

Privileged containers are not supported on Windows. Instead HostProcess containers can be used on Windows to perform many of the tasks performed by privileged containers on Linux.

8 - Controlling Access to the Kubernetes API

This page provides an overview of controlling access to the Kubernetes API.

Users access the Kubernetes API using `kubectl`, client libraries, or by making REST requests. Both human users and Kubernetes service accounts can be authorized for API access. When a request reaches the API, it goes through several stages, illustrated in the following diagram:



Transport security

By default, the Kubernetes API server listens on port 6443 on the first non-localhost network interface, protected by TLS. In a typical production Kubernetes cluster, the API serves on port 443. The port can be changed with the `--secure-port`, and the listening IP address with the `--bind-address` flag.

The API server presents a certificate. This certificate may be signed using a private certificate authority (CA), or based on a public key infrastructure linked to a generally recognized CA. The certificate and corresponding private key can be set by using the `--tls-cert-file` and `--tls-private-key-file` flags.

If your cluster uses a private certificate authority, you need a copy of that CA certificate configured into your `~/.kube/config` on the client, so that you can trust the connection and be confident it was not intercepted.

Your client can present a TLS client certificate at this stage.

Authentication

Once TLS is established, the HTTP request moves to the Authentication step. This is shown as step **1** in the diagram. The cluster creation script or cluster admin configures the API server to run one or more Authenticator modules. Authenticators are described in more detail in Authentication.

The input to the authentication step is the entire HTTP request; however, it typically examines the headers and/or client certificate.

Authentication modules include client certificates, password, and plain tokens, bootstrap tokens, and JSON Web Tokens (used for service accounts).

Multiple authentication modules can be specified, in which case each one is tried in sequence, until one of them succeeds.

If the request cannot be authenticated, it is rejected with HTTP status code 401. Otherwise, the user is authenticated as a specific `username`, and the user name is available to subsequent steps to use in their decisions. Some authenticators also provide the group memberships of the user, while other authenticators do not.

While Kubernetes uses usernames for access control decisions and in request logging, it does not have a `User` object nor does it store usernames or other information about users in its API.

Authorization

After the request is authenticated as coming from a specific user, the request must be authorized. This is shown as step **2** in the diagram.

A request must include the username of the requester, the requested action, and the object affected by the action. The request is authorized if an existing policy declares that the user has permissions to complete the requested action.

For example, if Bob has the policy below, then he can read pods only in the namespace `projectCaribou`:

```
{
    "apiVersion": "abac.authorization.kubernetes.io/v1beta1",
    "kind": "Policy",
    "spec": {
        "user": "bob",
        "namespace": "projectCaribou",
        "resource": "pods",
        "readonly": true
    }
}
```

If Bob makes the following request, the request is authorized because he is allowed to read objects in the `projectCaribou` namespace:

```
{
  "apiVersion": "authorization.k8s.io/v1beta1",
  "kind": "SubjectAccessReview",
  "spec": {
    "resourceAttributes": {
      "namespace": "projectCaribou",
      "verb": "get",
      "group": "unicorn.example.org",
      "resource": "pods"
    }
  }
}
```

If Bob makes a request to write (`create` or `update`) to the objects in the `projectCaribou` namespace, his authorization is denied. If Bob makes a request to read (`get`) objects in a different namespace such as `projectFish` , then his authorization is denied.

Kubernetes authorization requires that you use common REST attributes to interact with existing organization-wide or cloud-provider-wide access control systems. It is important to use REST formatting because these control systems might interact with other APIs besides the Kubernetes API.

Kubernetes supports multiple authorization modules, such as ABAC mode, RBAC Mode, and Webhook mode. When an administrator creates a cluster, they configure the authorization modules that should be used in the API server. If more than one authorization modules are configured, Kubernetes checks each module, and if any module authorizes the request, then the request can proceed. If all of the modules deny the request, then the request is denied (HTTP status code 403).

To learn more about Kubernetes authorization, including details about creating policies using the supported authorization modules, see Authorization.

Admission control

Admission Control modules are software modules that can modify or reject requests. In addition to the attributes available to Authorization modules, Admission Control modules can access the contents of the object that is being created or modified.

Admission controllers act on requests that create, modify, delete, or connect to (proxy) an object. Admission controllers do not act on requests that merely read objects. When multiple admission controllers are configured, they are called in order.

This is shown as step **3** in the diagram.

Unlike Authentication and Authorization modules, if any admission controller module rejects, then the request is immediately rejected.

In addition to rejecting objects, admission controllers can also set complex defaults for fields.

The available Admission Control modules are described in Admission Controllers.

Once a request passes all admission controllers, it is validated using the validation routines for the corresponding API object, and then written to the object store (shown as step **4**).

Auditing

Kubernetes auditing provides a security-relevant, chronological set of records documenting the sequence of actions in a cluster. The cluster audits the activities generated by users, by applications that use the Kubernetes API, and by the control plane itself.

For more information, see Auditing.

What's next

Read more documentation on authentication, authorization and API access control:

- Authenticating
  - Authenticating with Bootstrap Tokens
- Admission Controllers
  - Dynamic Admission Control
- Authorization
  - Role Based Access Control
  - Attribute Based Access Control
  - Node Authorization
  - Webhook Authorization
- Certificate Signing Requests
  - including CSR approval and certificate signing
- Service accounts
  - Developer guide
  - Administration

You can learn about:

- how Pods can use Secrets to obtain API credentials.

9 - Role Based Access Control Good Practices

Principles and practices for good RBAC design for cluster operators.

Kubernetes RBAC is a key security control to ensure that cluster users and workloads have only the access to resources required to execute their roles. It is important to ensure that, when designing permissions for cluster users, the cluster administrator understands the areas where privilege escalation could occur, to reduce the risk of excessive access leading to security incidents.

The good practices laid out here should be read in conjunction with the general RBAC documentation.

General good practice

Least privilege

Ideally, minimal RBAC rights should be assigned to users and service accounts. Only permissions explicitly required for their operation should be used. While each cluster will be different, some general rules that can be applied are :

- Assign permissions at the namespace level where possible. Use RoleBindings as opposed to ClusterRoleBindings to give users rights only within a specific namespace.
- Avoid providing wildcard permissions when possible, especially to all resources. As Kubernetes is an extensible system, providing wildcard access gives rights not just to all object types that currently exist in the cluster, but also to all object types which are created in the future.
- Administrators should not use `cluster-admin` accounts except where specifically needed. Providing a low privileged account with impersonation rights can avoid accidental modification of cluster resources.
- Avoid adding users to the `system:masters` group. Any user who is a member of this group bypasses all RBAC rights checks and will always have unrestricted superuser access, which cannot be revoked by removing RoleBindings or ClusterRoleBindings. As an aside, if a cluster is using an authorization webhook, membership of this group also bypasses that webhook (requests from users who are members of that group are never sent to the webhook)

Minimize distribution of privileged tokens

Ideally, pods shouldn't be assigned service accounts that have been granted powerful permissions (for example, any of the rights listed under privilege escalation risks). In cases where a workload requires powerful permissions, consider the following practices:

- Limit the number of nodes running powerful pods. Ensure that any DaemonSets you run are necessary and are run with least privilege to limit the blast radius of container escapes.
- Avoid running powerful pods alongside untrusted or publicly-exposed ones. Consider using Taints and Toleration, NodeAffinity, or PodAntiAffinity to ensure pods don't run alongside untrusted or less-trusted Pods. Pay special attention to situations where less-trustworthy Pods are not meeting the **Restricted** Pod Security Standard.

Hardening

Kubernetes defaults to providing access which may not be required in every cluster. Reviewing the RBAC rights provided by default can provide opportunities for security hardening. In general, changes should not be made to rights provided to `system:` accounts some options to harden cluster rights exist:

- Review bindings for the `system:unauthenticated` group and remove them where possible, as this gives access to anyone who can contact the API server at a network level.
- Avoid the default auto-mounting of service account tokens by setting `automountServiceAccountToken: false` . For more details, see using default service account token. Setting this value for a Pod will overwrite the service account setting, workloads which require service account tokens can still mount them.

Periodic review

It is vital to periodically review the Kubernetes RBAC settings for redundant entries and possible privilege escalations. If an attacker is able to create a user account with the same name as a deleted user, they can automatically inherit all the rights of the deleted user, especially the rights assigned to that user.

Kubernetes RBAC - privilege escalation risks

Within Kubernetes RBAC there are a number of privileges which, if granted, can allow a user or a service account to escalate their privileges in the cluster or affect systems

outside the cluster.

This section is intended to provide visibility of the areas where cluster operators should take care, to ensure that they do not inadvertently allow for more access to clusters than intended.

Listing secrets

It is generally clear that allowing `get` access on Secrets will allow a user to read their contents. It is also important to note that `list` and `watch` access also effectively allow for users to reveal the Secret contents. For example, when a List response is returned (for example, via `kubectl get secrets -A -o yaml`), the response includes the contents of all Secrets.

Workload creation

Permission to create workloads (either Pods, or workload resources that manage Pods) in a namespace implicitly grants access to many other resources in that namespace, such as Secrets, ConfigMaps, and PersistentVolumes that can be mounted in Pods. Additionally, since Pods can run as any ServiceAccount, granting permission to create workloads also implicitly grants the API access levels of any service account in that namespace.

Users who can run privileged Pods can use that access to gain node access and potentially to further elevate their privileges. Where you do not fully trust a user or other principal with the ability to create suitably secure and isolated Pods, you should enforce either the **Baseline** or **Restricted** Pod Security Standard. You can use Pod Security admission or other (third party) mechanisms to implement that enforcement.

For these reasons, namespaces should be used to separate resources requiring different levels of trust or tenancy. It is still considered best practice to follow least privilege principles and assign the minimum set of permissions, but boundaries within a namespace should be considered weak.

Persistent volume creation

If someone - or some application - is allowed to create arbitrary PersistentVolumes, that access includes the creation of `hostPath` volumes, which then means that a Pod would get access to the underlying host filesystem(s) on the associated node. Granting that ability is a security risk.

There are many ways a container with unrestricted access to the host filesystem can escalate privileges, including reading data from other containers, and abusing the credentials of system services, such as Kubelet.

You should only allow access to create PersistentVolume objects for:

- Users (cluster operators) that need this access for their work, and who you trust.
- The Kubernetes control plane components which creates PersistentVolumes based on PersistentVolumeClaims that are configured for automatic provisioning. This is usually setup by the Kubernetes provider or by the operator when installing a CSI driver.

Where access to persistent storage is required trusted administrators should create PersistentVolumes, and constrained users should use PersistentVolumeClaims to access that storage.

Access to `proxy` subresource of Nodes

Users with access to the `nodes/proxy` sub-resource have rights to the Kubelet API, which allows for command execution on every pod on the node(s) to which they have rights. This access bypasses audit logging and admission control, so care should be taken before granting any rights to this resource. These APIs can be exercised via websocket HTTP `GET` requests, which only requires authorization of the **get** verb. This means that **get** permission on `nodes/proxy` is not a read-only permission. For example, permission to **get** `nodes/proxy` provides access to privileged kubelet APIs that can retrieve container logs or execute and attach to pod processes, even when a caller does not have the equivalent permissions through the Kubernetes API.

See Kubelet authentication/authorization for more information.

Escalate verb

Generally, the RBAC system prevents users from creating clusterroles with more rights than the user possesses. The exception to this is the `escalate` verb. As noted in the RBAC documentation, users with this right can effectively escalate their privileges.

Bind verb

Similar to the `escalate` verb, granting users this right allows for the bypass of Kubernetes in-built protections against privilege escalation, allowing users to create bindings to roles with rights they do not already have.

Impersonate verb

This verb allows users to impersonate and gain the rights of other users in the cluster. Care should be taken when granting it, to ensure that excessive permissions cannot be gained via one of the impersonated accounts.

CSRs and certificate issuing

The CSR API allows for users with `create` rights to CSRs and `update` rights on `certificatesigningrequests/approval` where the signer is `kubernetes.io/kube-apiserver-client` to create new client certificates which allow users to authenticate to the cluster. Those client certificates can have arbitrary names including duplicates of Kubernetes system components. This will effectively allow for privilege escalation.

Token request

Users with `create` rights on `serviceaccounts/token` can create TokenRequests to issue tokens for existing service accounts.

Control admission webhooks

Users with control over `validatingwebhookconfigurations` or `mutatingwebhookconfigurations` can control webhooks that can read any object admitted to the cluster, and in the case of mutating webhooks, also mutate admitted objects.

Namespace modification

Users who can perform **patch** operations on Namespace objects (through a namespaced RoleBinding to a Role with that access) can modify labels on that namespace. In clusters where Pod Security Admission is used, this may allow a user to configure the namespace for a more permissive policy than intended by the administrators. For clusters where NetworkPolicy is used, users may be set labels that indirectly allow access to services that an administrator did not intend to allow.

Kubernetes RBAC - denial of service risks

Object creation denial-of-service

Users who have rights to create objects in a cluster may be able to create sufficient large objects to create a denial of service condition either based on the size or number of objects, as discussed in etcd used by Kubernetes is vulnerable to OOM attack. This may be specifically relevant in multi-tenant clusters if semi-trusted or untrusted users are allowed limited access to a system.

One option for mitigation of this issue would be to use resource quotas to limit the quantity of objects which can be created.

What's next

- To learn more about RBAC, see the RBAC documentation.

10 - Good practices for Kubernetes Secrets

Principles and practices for good Secret management for cluster administrators and application developers.

In Kubernetes, a Secret is an object that stores sensitive information, such as passwords, OAuth tokens, and SSH keys.

Secrets give you more control over how sensitive information is used and reduces the risk of accidental exposure. Secret values are encoded as base64 strings and are stored unencrypted by default, but can be configured to be encrypted at rest.

A Pod can reference the Secret in a variety of ways, such as in a volume mount or as an environment variable. Secrets are designed for confidential data and ConfigMaps are designed for non-confidential data.

The following good practices are intended for both cluster administrators and application developers. Use these guidelines to improve the security of your sensitive information in Secret objects, as well as to more effectively manage your Secrets.

Cluster administrators

This section provides good practices that cluster administrators can use to improve the security of confidential information in the cluster.

Configure encryption at rest

By default, Secret objects are stored unencrypted in etcd. You should configure encryption of your Secret data in `etcd` . For instructions, refer to Encrypt Secret Data at Rest.

Configure least-privilege access to Secrets

When planning your access control mechanism, such as Kubernetes Role-based Access Control (RBAC), consider the following guidelines for access to `Secret` objects. You should also follow the other guidelines in RBAC good practices.

- **Components**: Restrict `watch` or `list` access to only the most privileged, system-level components. Only grant `get` access for Secrets if the component's normal behavior

requires it.

- **Humans**: Restrict `get`, `watch`, or `list` access to Secrets. Only allow cluster administrators to access `etcd`. This includes read-only access. For more complex access control, such as restricting access to Secrets with specific annotations, consider using third-party authorization mechanisms.

> **Caution:**
>
> Granting `list` access to Secrets implicitly lets the subject fetch the contents of the Secrets.

A user who can create a Pod that uses a Secret can also see the value of that Secret. Even if cluster policies do not allow a user to read the Secret directly, the same user could have access to run a Pod that then exposes the Secret. You can detect or limit the impact caused by Secret data being exposed, either intentionally or unintentionally, by a user with this access. Some recommendations include:

- Use short-lived Secrets
- Implement audit rules that alert on specific events, such as concurrent reading of multiple Secrets by a single user

Restrict Access for Secrets

Use separate namespaces to isolate access to mounted secrets.

Improve etcd management policies

Consider wiping or shredding the durable storage used by `etcd` once it is no longer in use.

If there are multiple `etcd` instances, configure encrypted SSL/TLS communication between the instances to protect the Secret data in transit.

Configure access to external Secrets

> **Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects,

> which are listed alphabetically. To add a project to this list, read the content guide before submitting a change. More information.

You can use third-party Secrets store providers to keep your confidential data outside your cluster and then configure Pods to access that information. The Kubernetes Secrets Store CSI Driver is a DaemonSet that lets the kubelet retrieve Secrets from external stores, and mount the Secrets as a volume into specific Pods that you authorize to access the data.

For a list of supported providers, refer to Providers for the Secret Store CSI Driver.

Good practices for using swap memory

For best practices for setting swap memory for Linux nodes, please refer to swap memory management.

Developers

This section provides good practices for developers to use to improve the security of confidential data when building and deploying Kubernetes resources.

Restrict Secret access to specific containers

If you are defining multiple containers in a Pod, and only one of those containers needs access to a Secret, define the volume mount or environment variable configuration so that the other containers do not have access to that Secret.

Protect Secret data after reading

Applications still need to protect the value of confidential information after reading it from an environment variable or volume. For example, your application must avoid logging the secret data in the clear or transmitting it to an untrusted party.

Avoid sharing Secret manifests

If you configure a Secret through a manifest, with the secret data encoded as base64, sharing this file or checking it in to a source repository means the secret is available to

everyone who can read the manifest.

> **Caution:**
>
> Base64 encoding is *not* an encryption method, it provides no additional confidentiality over plain text.

11 - Multi-tenancy

This page provides an overview of available configuration options and best practices for cluster multi-tenancy.

Sharing clusters saves costs and simplifies administration. However, sharing clusters also presents challenges such as security, fairness, and managing *noisy neighbors*.

Clusters can be shared in many ways. In some cases, different applications may run in the same cluster. In other cases, multiple instances of the same application may run in the same cluster, one for each end user. All these types of sharing are frequently described using the umbrella term *multi-tenancy*.

While Kubernetes does not have first-class concepts of end users or tenants, it provides several features to help manage different tenancy requirements. These are discussed below.

Use cases

The first step to determining how to share your cluster is understanding your use case, so you can evaluate the patterns and tools available. In general, multi-tenancy in Kubernetes clusters falls into two broad categories, though many variations and hybrids are also possible.

Multiple teams

A common form of multi-tenancy is to share a cluster between multiple teams within an organization, each of whom may operate one or more workloads. These workloads frequently need to communicate with each other, and with other workloads located on the same or different clusters.

In this scenario, members of the teams often have direct access to Kubernetes resources via tools such as `kubectl`, or indirect access through GitOps controllers or other types of release automation tools. There is often some level of trust between members of different teams, but Kubernetes policies such as RBAC, quotas, and network policies are essential to safely and fairly share clusters.

Multiple customers

The other major form of multi-tenancy frequently involves a Software-as-a-Service (SaaS) vendor running multiple instances of a workload for customers. This business model is so strongly associated with this deployment style that many people call it "SaaS tenancy." However, a better term might be "multi-customer tenancy," since SaaS vendors may also use other deployment models, and this deployment model can also be used outside of SaaS.

In this scenario, the customers do not have access to the cluster; Kubernetes is invisible from their perspective and is only used by the vendor to manage the workloads. Cost optimization is frequently a critical concern, and Kubernetes policies are used to ensure that the workloads are strongly isolated from each other.

Terminology

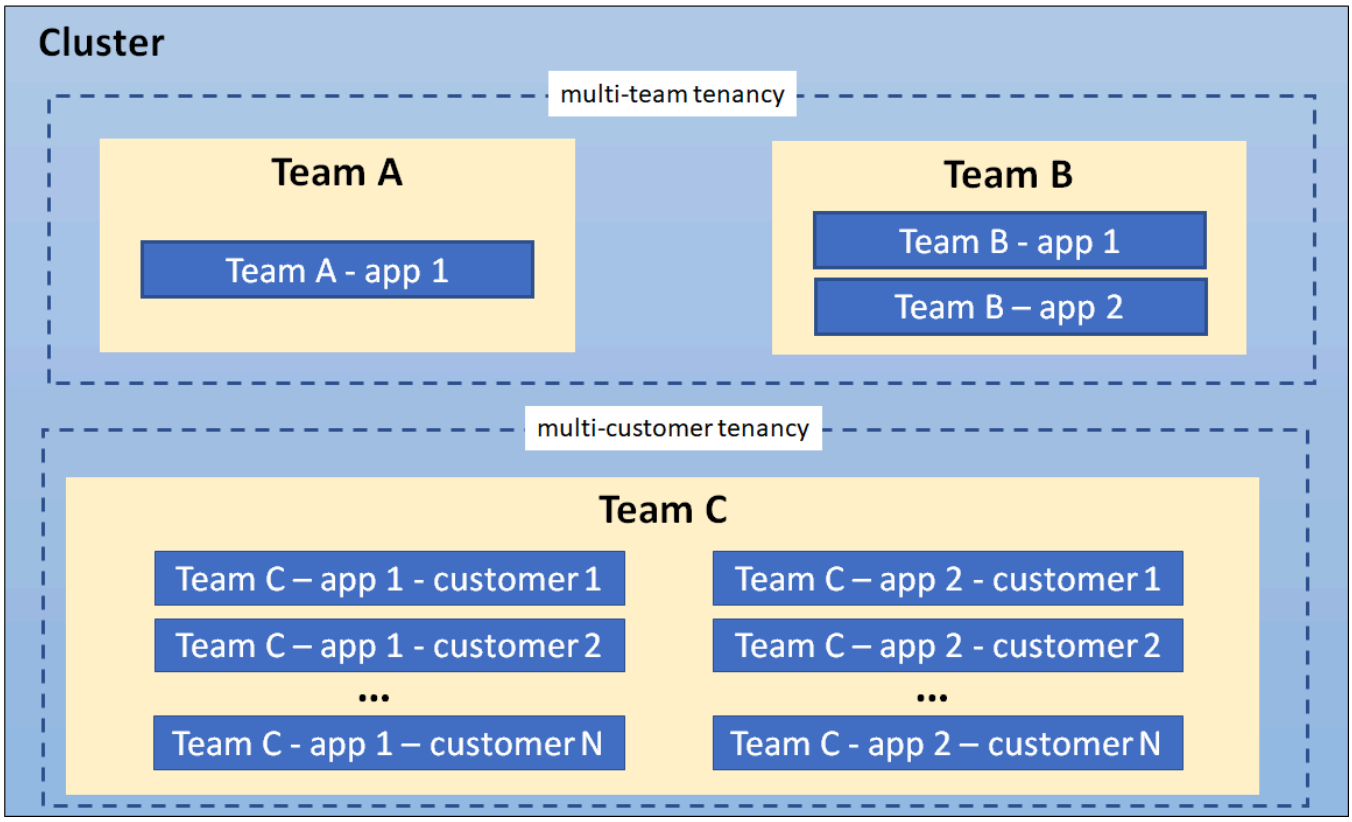
Tenants

When discussing multi-tenancy in Kubernetes, there is no single definition for a "tenant". Rather, the definition of a tenant will vary depending on whether multi-team or multi-customer tenancy is being discussed.

In multi-team usage, a tenant is typically a team, where each team typically deploys a small number of workloads that scales with the complexity of the service. However, the definition of "team" may itself be fuzzy, as teams may be organized into higher-level divisions or subdivided into smaller teams.

By contrast, if each team deploys dedicated workloads for each new client, they are using a multi-customer model of tenancy. In this case, a "tenant" is simply a group of users who share a single workload. This may be as large as an entire company, or as small as a single team at that company.

In many cases, the same organization may use both definitions of "tenants" in different contexts. For example, a platform team may offer shared services such as security tools and databases to multiple internal “customers” and a SaaS vendor may also have multiple teams sharing a development cluster. Finally, hybrid architectures are also possible, such as a SaaS provider using a combination of per-customer workloads for sensitive data, combined with multi-tenant shared services.



A cluster showing coexisting tenancy models

Isolation

There are several ways to design and build multi-tenant solutions with Kubernetes. Each of these methods comes with its own set of tradeoffs that impact the isolation level, implementation effort, operational complexity, and cost of service.

A Kubernetes cluster consists of a control plane which runs Kubernetes software, and a data plane consisting of worker nodes where tenant workloads are executed as pods. Tenant isolation can be applied in both the control plane and the data plane based on organizational requirements.

The level of isolation offered is sometimes described using terms like “hard” multi-tenancy, which implies strong isolation, and “soft” multi-tenancy, which implies weaker isolation. In particular, "hard" multi-tenancy is often used to describe cases where the tenants do not trust each other, often from security and resource sharing perspectives (e.g. guarding against attacks such as data exfiltration or DoS). Since data planes typically have much larger attack surfaces, "hard" multi-tenancy often requires extra attention to isolating the data-plane, though control plane isolation also remains critical.

However, the terms "hard" and "soft" can often be confusing, as there is no single definition that will apply to all users. Rather, "hardness" or "softness" is better understood

as a broad spectrum, with many different techniques that can be used to maintain different types of isolation in your clusters, based on your requirements.

In more extreme cases, it may be easier or necessary to forgo any cluster-level sharing at all and assign each tenant their dedicated cluster, possibly even running on dedicated hardware if VMs are not considered an adequate security boundary. This may be easier with managed Kubernetes clusters, where the overhead of creating and operating clusters is at least somewhat taken on by a cloud provider. The benefit of stronger tenant isolation must be evaluated against the cost and complexity of managing multiple clusters. The Multi-cluster SIG is responsible for addressing these types of use cases.

The remainder of this page focuses on isolation techniques used for shared Kubernetes clusters. However, even if you are considering dedicated clusters, it may be valuable to review these recommendations, as it will give you the flexibility to shift to shared clusters in the future if your needs or capabilities change.

Control plane isolation

Control plane isolation ensures that different tenants cannot access or affect each others' Kubernetes API resources.

Namespaces

In Kubernetes, a Namespace provides a mechanism for isolating groups of API resources within a single cluster. This isolation has two key dimensions:

1. Object names within a namespace can overlap with names in other namespaces, similar to files in folders. This allows tenants to name their resources without having to consider what other tenants are doing.

2. Many Kubernetes security policies are scoped to namespaces. For example, RBAC Roles and Network Policies are namespace-scoped resources. Using RBAC, Users and Service Accounts can be restricted to a namespace.

In a multi-tenant environment, a Namespace helps segment a tenant's workload into a logical and distinct management unit. In fact, a common practice is to isolate every workload in its own namespace, even if multiple workloads are operated by the same tenant. This ensures that each workload has its own identity and can be configured with an appropriate security policy.

The namespace isolation model requires configuration of several other Kubernetes resources, networking plugins, and adherence to security best practices to properly isolate tenant workloads. These considerations are discussed below.

Access controls

The most important type of isolation for the control plane is authorization. If teams or their workloads can access or modify each others' API resources, they can change or disable all other types of policies thereby negating any protection those policies may offer. As a result, it is critical to ensure that each tenant has the appropriate access to only the namespaces they need, and no more. This is known as the "Principle of Least Privilege."

Role-based access control (RBAC) is commonly used to enforce authorization in the Kubernetes control plane, for both users and workloads (service accounts). Roles and RoleBindings are Kubernetes objects that are used at a namespace level to enforce access control in your application; similar objects exist for authorizing access to cluster-level objects, though these are less useful for multi-tenant clusters.

In a multi-team environment, RBAC must be used to restrict tenants' access to the appropriate namespaces, and ensure that cluster-wide resources can only be accessed or modified by privileged users such as cluster administrators.

If a policy ends up granting a user more permissions than they need, this is likely a signal that the namespace containing the affected resources should be refactored into finer-grained namespaces. Namespace management tools may simplify the management of these finer-grained namespaces by applying common RBAC policies to different namespaces, while still allowing fine-grained policies where necessary.

Quotas

Kubernetes workloads consume node resources, like CPU and memory. In a multi-tenant environment, you can use Resource Quotas to manage resource usage of tenant workloads. For the multiple teams use case, where tenants have access to the Kubernetes API, you can use resource quotas to limit the number of API resources (for example: the number of Pods, or the number of ConfigMaps) that a tenant can create. Limits on object count ensure fairness and aim to avoid *noisy neighbor* issues from affecting other tenants that share a control plane.

Resource quotas are namespaced objects. By mapping tenants to namespaces, cluster admins can use quotas to ensure that a tenant cannot monopolize a cluster's resources or overwhelm its control plane. Namespace management tools simplify the administration of

quotas. In addition, while Kubernetes quotas only apply within a single namespace, some namespace management tools allow groups of namespaces to share quotas, giving administrators far more flexibility with less effort than built-in quotas.

Quotas prevent a single tenant from consuming greater than their allocated share of resources hence minimizing the “noisy neighbor” issue, where one tenant negatively impacts the performance of other tenants' workloads.

When you apply a quota to namespace, Kubernetes requires you to also specify resource requests and limits for each container. Limits are the upper bound for the amount of resources that a container can consume. Containers that attempt to consume resources that exceed the configured limits will either be throttled or killed, based on the resource type. When resource requests are set lower than limits, each container is guaranteed the requested amount but there may still be some potential for impact across workloads.

Quotas cannot protect against all kinds of resource sharing, such as network traffic. Node isolation (described below) may be a better solution for this problem.

Data Plane Isolation

Data plane isolation ensures that pods and workloads for different tenants are sufficiently isolated.

Network isolation

By default, all pods in a Kubernetes cluster are allowed to communicate with each other, and all network traffic is unencrypted. This can lead to security vulnerabilities where traffic is accidentally or maliciously sent to an unintended destination, or is intercepted by a workload on a compromised node.

Pod-to-pod communication can be controlled using Network Policies, which restrict communication between pods using namespace labels or IP address ranges. In a multi-tenant environment where strict network isolation between tenants is required, starting with a default policy that denies communication between pods is recommended with another rule that allows all pods to query the DNS server for name resolution. With such a default policy in place, you can begin adding more permissive rules that allow for communication within a namespace. It is also recommended not to use empty label selector '{}' for namespaceSelector field in network policy definition, in case traffic need to be allowed between namespaces. This scheme can be further refined as required. Note that this only applies to pods within a single control plane; pods that belong to different virtual control planes cannot talk to each other via Kubernetes networking.

Namespace management tools may simplify the creation of default or common network policies. In addition, some of these tools allow you to enforce a consistent set of namespace labels across your cluster, ensuring that they are a trusted basis for your policies.

> **Warning:**
>
> Network policies require a CNI plugin that supports the implementation of network policies. Otherwise, NetworkPolicy resources will be ignored.

More advanced network isolation may be provided by service meshes, which provide OSI Layer 7 policies based on workload identity, in addition to namespaces. These higher-level policies can make it easier to manage namespace-based multi-tenancy, especially when multiple namespaces are dedicated to a single tenant. They frequently also offer encryption using mutual TLS, protecting your data even in the presence of a compromised node, and work across dedicated or virtual clusters. However, they can be significantly more complex to manage and may not be appropriate for all users.

Storage isolation

Kubernetes offers several types of volumes that can be used as persistent storage for workloads. For security and data-isolation, dynamic volume provisioning is recommended and volume types that use node resources should be avoided.

StorageClasses allow you to describe custom "classes" of storage offered by your cluster, based on quality-of-service levels, backup policies, or custom policies determined by the cluster administrators.

Pods can request storage using a PersistentVolumeClaim. A PersistentVolumeClaim is a namespaced resource, which enables isolating portions of the storage system and dedicating it to tenants within the shared Kubernetes cluster. However, it is important to note that a PersistentVolume is a cluster-wide resource and has a lifecycle independent of workloads and namespaces.

For example, you can configure a separate StorageClass for each tenant and use this to strengthen isolation. If a StorageClass is shared, you should set a reclaim policy of `Delete` to ensure that a PersistentVolume cannot be reused across different namespaces.

Sandboxing containers

Kubernetes pods are composed of one or more containers that execute on worker nodes. Containers utilize OS-level virtualization and hence offer a weaker isolation boundary than virtual machines that utilize hardware-based virtualization.

In a shared environment, unpatched vulnerabilities in the application and system layers can be exploited by attackers for container breakouts and remote code execution that allow access to host resources. In some applications, like a Content Management System (CMS), customers may be allowed the ability to upload and execute untrusted scripts or code. In either case, mechanisms to further isolate and protect workloads using strong isolation are desirable.

Sandboxing provides a way to isolate workloads running in a shared cluster. It typically involves running each pod in a separate execution environment such as a virtual machine or a userspace kernel. Sandboxing is often recommended when you are running untrusted code, where workloads are assumed to be malicious. Part of the reason this type of isolation is necessary is because containers are processes running on a shared kernel; they mount file systems like `/sys` and `/proc` from the underlying host, making them less secure than an application that runs on a virtual machine which has its own kernel. While controls such as seccomp, AppArmor, and SELinux can be used to strengthen the security of containers, it is hard to apply a universal set of rules to all workloads running in a shared cluster. Running workloads in a sandbox environment helps to insulate the host from container escapes, where an attacker exploits a vulnerability to gain access to the host system and all the processes/files running on that host.

Virtual machines and userspace kernels are two popular approaches to sandboxing.

Node Isolation

Node isolation is another technique that you can use to isolate tenant workloads from each other. With node isolation, a set of nodes is dedicated to running pods from a particular tenant and co-mingling of tenant pods is prohibited. This configuration reduces the noisy tenant issue, as all pods running on a node will belong to a single tenant. The risk of information disclosure is slightly lower with node isolation because an attacker that manages to escape from a container will only have access to the containers and volumes mounted to that node.

Although workloads from different tenants are running on different nodes, it is important to be aware that the kubelet and (unless using virtual control planes) the API service are still shared services. A skilled attacker could use the permissions assigned to the kubelet or other pods running on the node to move laterally within the cluster and gain access to tenant workloads running on other nodes. If this is a major concern, consider

implementing compensating controls such as seccomp, AppArmor or SELinux or explore using sandboxed containers or creating separate clusters for each tenant.

Node isolation is a little easier to reason about from a billing standpoint than sandboxing containers since you can charge back per node rather than per pod. It also has fewer compatibility and performance issues and may be easier to implement than sandboxing containers. For example, nodes for each tenant can be configured with taints so that only pods with the corresponding toleration can run on them. A mutating webhook could then be used to automatically add tolerations and node affinities to pods deployed into tenant namespaces so that they run on a specific set of nodes designated for that tenant.

Node isolation can be implemented using pod node selectors.

Additional Considerations

This section discusses other Kubernetes constructs and patterns that are relevant for multi-tenancy.

API Priority and Fairness

API priority and fairness is a Kubernetes feature that allows you to assign a priority to certain pods running within the cluster. When an application calls the Kubernetes API, the API server evaluates the priority assigned to pod. Calls from pods with higher priority are fulfilled before those with a lower priority. When contention is high, lower priority calls can be queued until the server is less busy or you can reject the requests.

Using API priority and fairness will not be very common in SaaS environments unless you are allowing customers to run applications that interface with the Kubernetes API, for example, a controller.

Quality-of-Service (QoS)

When you're running a SaaS application, you may want the ability to offer different Quality-of-Service (QoS) tiers of service to different tenants. For example, you may have freemium service that comes with fewer performance guarantees and features and a for-fee service tier with specific performance guarantees. Fortunately, there are several Kubernetes constructs that can help you accomplish this within a shared cluster, including network QoS, storage classes, and pod priority and preemption. The idea with each of these is to provide tenants with the quality of service that they paid for. Let's start by looking at networking QoS.

Typically, all pods on a node share a network interface. Without network QoS, some pods may consume an unfair share of the available bandwidth at the expense of other pods. The Kubernetes bandwidth plugin creates an extended resource for networking that allows you to use Kubernetes resources constructs, i.e. requests/limits, to apply rate limits to pods by using Linux tc queues. Be aware that the plugin is considered experimental as per the Network Plugins documentation and should be thoroughly tested before use in production environments.

For storage QoS, you will likely want to create different storage classes or profiles with different performance characteristics. Each storage profile can be associated with a different tier of service that is optimized for different workloads such IO, redundancy, or throughput. Additional logic might be necessary to allow the tenant to associate the appropriate storage profile with their workload.

Finally, there's pod priority and preemption where you can assign priority values to pods. When scheduling pods, the scheduler will try evicting pods with lower priority when there are insufficient resources to schedule pods that are assigned a higher priority. If you have a use case where tenants have different service tiers in a shared cluster e.g. free and paid, you may want to give higher priority to certain tiers using this feature.

DNS

Kubernetes clusters include a Domain Name System (DNS) service to provide translations from names to IP addresses, for all Services and Pods. By default, the Kubernetes DNS service allows lookups across all namespaces in the cluster.

In multi-tenant environments where tenants can access pods and other Kubernetes resources, or where stronger isolation is required, it may be necessary to prevent pods from looking up services in other Namespaces. You can restrict cross-namespace DNS lookups by configuring security rules for the DNS service. For example, CoreDNS (the default DNS service for Kubernetes) can leverage Kubernetes metadata to restrict queries to Pods and Services within a namespace. For more information, read an example of configuring this within the CoreDNS documentation.

When a Virtual Control Plane per tenant model is used, a DNS service must be configured per tenant or a multi-tenant DNS service must be used. Here is an example of a customized version of CoreDNS that supports multiple tenants.

Operators

Operators are Kubernetes controllers that manage applications. Operators can simplify the management of multiple instances of an application, like a database service, which makes them a common building block in the multi-consumer (SaaS) multi-tenancy use case.

Operators used in a multi-tenant environment should follow a stricter set of guidelines. Specifically, the Operator should:

- Support creating resources within different tenant namespaces, rather than just in the namespace in which the Operator is deployed.
- Ensure that the Pods are configured with resource requests and limits, to ensure scheduling and fairness.
- Support configuration of Pods for data-plane isolation techniques such as node isolation and sandboxed containers.

Implementations

There are two primary ways to share a Kubernetes cluster for multi-tenancy: using Namespaces (that is, a Namespace per tenant) or by virtualizing the control plane (that is, virtual control plane per tenant).

In both cases, data plane isolation, and management of additional considerations such as API Priority and Fairness, is also recommended.

Namespace isolation is well-supported by Kubernetes, has a negligible resource cost, and provides mechanisms to allow tenants to interact appropriately, such as by allowing service-to-service communication. However, it can be difficult to configure, and doesn't apply to Kubernetes resources that can't be namespaced, such as Custom Resource Definitions, Storage Classes, and Webhooks.

Control plane virtualization allows for isolation of non-namespaced resources at the cost of somewhat higher resource usage and more difficult cross-tenant sharing. It is a good option when namespace isolation is insufficient but dedicated clusters are undesirable, due to the high cost of maintaining them (especially on-prem) or due to their higher overhead and lack of resource sharing. However, even within a virtualized control plane, you will likely see benefits by using namespaces as well.

The two options are discussed in more detail in the following sections.

Namespace per tenant

As previously mentioned, you should consider isolating each workload in its own namespace, even if you are using dedicated clusters or virtualized control planes. This ensures that each workload only has access to its own resources, such as ConfigMaps and Secrets, and allows you to tailor dedicated security policies for each workload. In addition, it is a best practice to give each namespace names that are unique across your entire fleet (that is, even if they are in separate clusters), as this gives you the flexibility to switch between dedicated and shared clusters in the future, or to use multi-cluster tooling such as service meshes.

Conversely, there are also advantages to assigning namespaces at the tenant level, not just the workload level, since there are often policies that apply to all workloads owned by a single tenant. However, this raises its own problems. Firstly, this makes it difficult or impossible to customize policies to individual workloads, and secondly, it may be challenging to come up with a single level of "tenancy" that should be given a namespace. For example, an organization may have divisions, teams, and subteams - which should be assigned a namespace?

One possible approach is to organize your namespaces into hierarchies, and share certain policies and resources between them. This could include managing namespace labels, namespace lifecycles, delegated access, and shared resource quotas across related namespaces. These capabilities can be useful in both multi-team and multi-customer scenarios.

Virtual control plane per tenant

Another form of control-plane isolation is to use Kubernetes extensions to provide each tenant a virtual control-plane that enables segmentation of cluster-wide API resources. Data plane isolation techniques can be used with this model to securely manage worker nodes across tenants.

The virtual control plane based multi-tenancy model extends namespace-based multi-tenancy by providing each tenant with dedicated control plane components, and hence complete control over cluster-wide resources and add-on services. Worker nodes are shared across all tenants, and are managed by a Kubernetes cluster that is normally inaccessible to tenants. This cluster is often referred to as a *super-cluster* (or sometimes as a *host-cluster*). Since a tenant's control-plane is not directly associated with underlying compute resources it is referred to as a *virtual control plane*.

A virtual control plane typically consists of the Kubernetes API server, the controller manager, and the etcd data store. It interacts with the super cluster via a metadata synchronization controller which coordinates changes across tenant control planes and the control plane of the super-cluster.

By using per-tenant dedicated control planes, most of the isolation problems due to sharing one API server among all tenants are solved. Examples include noisy neighbors in the control plane, uncontrollable blast radius of policy misconfigurations, and conflicts between cluster scope objects such as webhooks and CRDs. Hence, the virtual control plane model is particularly suitable for cases where each tenant requires access to a Kubernetes API server and expects the full cluster manageability.

The improved isolation comes at the cost of running and maintaining an individual virtual control plane per tenant. In addition, per-tenant control planes do not solve isolation problems in the data plane, such as node-level noisy neighbors or security threats. These must still be addressed separately.

12 - Hardening Guide - Authentication Mechanisms

Information on authentication options in Kubernetes and their security properties.

Selecting the appropriate authentication mechanism(s) is a crucial aspect of securing your cluster. Kubernetes provides several built-in mechanisms, each with its own strengths and weaknesses that should be carefully considered when choosing the best authentication mechanism for your cluster.

In general, it is recommended to enable as few authentication mechanisms as possible to simplify user management and prevent cases where users retain access to a cluster that is no longer required.

It is important to note that Kubernetes does not have an in-built user database within the cluster. Instead, it takes user information from the configured authentication system and uses that to make authorization decisions. Therefore, to audit user access, you need to review credentials from every configured authentication source.

For production clusters with multiple users directly accessing the Kubernetes API, it is recommended to use external authentication sources such as OIDC. The internal authentication mechanisms, such as client certificates and service account tokens, described below, are not suitable for this use case.

X.509 client certificate authentication

Kubernetes leverages X.509 client certificate authentication for system components, such as when the kubelet authenticates to the API Server. While this mechanism can also be used for user authentication, it might not be suitable for production use due to several restrictions:

- Client certificates cannot be individually revoked. Once compromised, a certificate can be used by an attacker until it expires. To mitigate this risk, it is recommended to configure short lifetimes for user authentication credentials created using client certificates.
- If a certificate needs to be invalidated, the certificate authority must be re-keyed, which can introduce availability risks to the cluster.

- There is no permanent record of client certificates created in the cluster. Therefore, all issued certificates must be recorded if you need to keep track of them.
- Private keys used for client certificate authentication cannot be password-protected. Anyone who can read the file containing the key will be able to make use of it.
- Using client certificate authentication requires a direct connection from the client to the API server without any intervening TLS termination points, which can complicate network architectures.
- Group data is embedded in the `O` value of the client certificate, which means the user's group memberships cannot be changed for the lifetime of the certificate.

Static token file

Although Kubernetes allows you to load credentials from a static token file located on the control plane node disks, this approach is not recommended for production servers due to several reasons:

- Credentials are stored in clear text on control plane node disks, which can be a security risk.
- Changing any credential requires a restart of the API server process to take effect, which can impact availability.
- There is no mechanism available to allow users to rotate their credentials. To rotate a credential, a cluster administrator must modify the token on disk and distribute it to the users.
- There is no lockout mechanism available to prevent brute-force attacks.

Bootstrap tokens

Bootstrap tokens are used for joining nodes to clusters and are not recommended for user authentication due to several reasons:

- They have hard-coded group memberships that are not suitable for general use, making them unsuitable for authentication purposes.
- Manually generating bootstrap tokens can lead to weak tokens that can be guessed by an attacker, which can be a security risk.
- There is no lockout mechanism available to prevent brute-force attacks, making it easier for attackers to guess or crack the token.

ServiceAccount secret tokens

Service account secrets are available as an option to allow workloads running in the cluster to authenticate to the API server. In Kubernetes < 1.23, these were the default option, however, they are being replaced with TokenRequest API tokens. While these secrets could be used for user authentication, they are generally unsuitable for a number of reasons:

- They cannot be set with an expiry and will remain valid until the associated service account is deleted.
- The authentication tokens are visible to any cluster user who can read secrets in the namespace that they are defined in.
- Service accounts cannot be added to arbitrary groups complicating RBAC management where they are used.

TokenRequest API tokens

The TokenRequest API is a useful tool for generating short-lived credentials for service authentication to the API server or third-party systems. However, it is not generally recommended for user authentication as there is no revocation method available, and distributing credentials to users in a secure manner can be challenging.

When using TokenRequest tokens for service authentication, it is recommended to implement a short lifespan to reduce the impact of compromised tokens.

OpenID Connect token authentication

Kubernetes supports integrating external authentication services with the Kubernetes API using OpenID Connect (OIDC). There is a wide variety of software that can be used to integrate Kubernetes with an identity provider. However, when using OIDC authentication in Kubernetes, it is important to consider the following hardening measures:

- The software installed in the cluster to support OIDC authentication should be isolated from general workloads as it will run with high privileges.
- Some Kubernetes managed services are limited in the OIDC providers that can be used.
- As with TokenRequest tokens, OIDC tokens should have a short lifespan to reduce the impact of compromised tokens.

Webhook token authentication

Webhook token authentication is another option for integrating external authentication providers into Kubernetes. This mechanism allows for an authentication service, either running inside the cluster or externally, to be contacted for an authentication decision over a webhook. It is important to note that the suitability of this mechanism will likely depend on the software used for the authentication service, and there are some Kubernetes-specific considerations to take into account.

To configure Webhook authentication, access to control plane server filesystems is required. This means that it will not be possible with Managed Kubernetes unless the provider specifically makes it available. Additionally, any software installed in the cluster to support this access should be isolated from general workloads, as it will run with high privileges.

Authenticating proxy

Another option for integrating external authentication systems into Kubernetes is to use an authenticating proxy. With this mechanism, Kubernetes expects to receive requests from the proxy with specific header values set, indicating the username and group memberships to assign for authorization purposes. It is important to note that there are specific considerations to take into account when using this mechanism.

Firstly, securely configured TLS must be used between the proxy and Kubernetes API server to mitigate the risk of traffic interception or sniffing attacks. This ensures that the communication between the proxy and Kubernetes API server is secure.

Secondly, it is important to be aware that an attacker who is able to modify the headers of the request may be able to gain unauthorized access to Kubernetes resources. As such, it is important to ensure that the headers are properly secured and cannot be tampered with.

What's next

- User Authentication
- Authenticating with Bootstrap Tokens
- kubelet Authentication
- Authenticating with Service Account Tokens

13 - Hardening Guide - Dynamic Resource Allocation

Information about hardening Dynamic Resource Allocation (DRA) authorization and access patterns.

Dynamic Resource Allocation (DRA) adds powerful scheduling and device management capabilities. Because DRA components update `ResourceClaim` status, cluster administrators should configure authorization for those updates with explicit, least-privilege RBAC.

ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

Starting in Kubernetes v1.36, DRA status updates use synthetic subresources and, in some cases, specialized node-aware verbs.

Harden DRA status update permissions

For DRA status updates,In addition to granting `update` permissions on the `resourceclaims/status` subresource, cluster administrators must grant permissions on specific "synthetic" subresources based on the exact fields a component needs to modify. This enforces the principle of least privilege between the scheduler, custom controllers, and DRA drivers.

The DRA authorization checks are divided into two synthetic subresources:

- **`resourceclaims/binding`**
  - Required to modify `status.allocation` and `status.reservedFor`.
  - Typically granted to the kube-scheduler and custom allocation controllers.
  - Uses standard `update` and `patch` verbs.
- **`resourceclaims/driver`**
  - Required to modify `status.devices`.
  - This check is performed per-driver to drivers from tampering with devices on different nodes and/or from other drivers.
  - Uses node-aware verbs for stricter scope.

Node-aware DRA verbs

When authorizing updates to `resourceclaims/driver`, use the appropriate specialized verb prefix:

- **`associated-node:<verb>`** (for example, `associated-node:update`)
  - For node-local drivers.
  - The API server verifies node association for the requesting driver.
- **`arbitrary-node:<verb>`** (for example, `arbitrary-node:patch`)
  - For control-plane or multi-node controllers that may update claims from any node.

Example RBAC patterns

Scheduler and allocation controller permissions

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: dra-binding-updater
rules:
  - apiGroups: ["resource.k8s.io"]
    resources: ["resourceclaims/status"]
    verbs: ["get", "patch", "update"]
  - apiGroups: ["resource.k8s.io"]
    resources: ["resourceclaims/binding"]
    verbs: ["patch", "update"]
```

Node-local DRA driver permissions

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: dra-node-driver-status-updater
rules:
  - apiGroups: ["resource.k8s.io"]
    resources: ["resourceclaims/status"]
```

```
    verbs: ["get", "patch", "update"]
  - apiGroups: ["resource.k8s.io"]
    resources: ["resourceclaims/driver"]
    verbs: ["associated-node:patch", "associated-node:update"]
    resourceNames: ["dra.example.com"]
```

Multi-node status controller permissions

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: dra-multinode-status-updater
rules:
  - apiGroups: ["resource.k8s.io"]
    resources: ["resourceclaims/status"]
    verbs: ["get", "patch", "update"]
  - apiGroups: ["resource.k8s.io"]
    resources: ["resourceclaims/driver"]
    verbs: ["arbitrary-node:patch", "arbitrary-node:update"]
    resourceNames: ["dra.example.com"]
```

Related cluster administrator task

To apply these patterns in a running cluster, see Harden Dynamic Resource Allocation in Your Cluster.

What's next

- Authorization
- Set Up DRA in a Cluster
- Dynamic Resource Allocation

14 - Hardening Guide - Scheduler Configuration

Information about how to make the Kubernetes scheduler more secure.

The Kubernetes scheduler is one of the critical components of the control plane.

This document covers how to improve the security posture of the Scheduler.

A misconfigured scheduler can have security implications. Such a scheduler can target specific nodes and evict the workloads or applications that are sharing the node and its resources. This can aid an attacker with a Yo-Yo attack: an attack on a vulnerable autoscaler.

kube-scheduler configuration

Scheduler authentication & authorization command line options

When setting up authentication configuration, it should be made sure that kube-scheduler's authentication remains consistent with kube-api-server's authentication. If any request has missing authentication headers, the authentication should happen through the kube-api-server allowing all authentication to be consistent in the cluster.

- `authentication-kubeconfig` : Make sure to provide a proper kubeconfig so that the scheduler can retrieve authentication configuration options from the API Server. This kubeconfig file should be protected with strict file permissions.
- `authentication-tolerate-lookup-failure` : Set this to `false` to make sure the scheduler *always* looks up its authentication configuration from the API server.
- `authentication-skip-lookup` : Set this to `false` to make sure the scheduler *always* looks up its authentication configuration from the API server.
- `authorization-always-allow-paths` : These paths should respond with data that is appropriate for anonymous authorization. Defaults to `/healthz,/readyz,/livez` .
- `profiling` : Set to `false` to disable the profiling endpoints which are provide debugging information but which should not be enabled on production clusters as they present a risk of denial of service or information leakage. The `--profiling` argument is deprecated and can now be provided through the KubeScheduler

DebuggingConfiguration. Profiling can be disabled through the kube-scheduler config by setting `enableProfiling` to `false`.

- `requestheader-client-ca-file`: Avoid passing this argument.

Scheduler networking command line options

- `bind-address`: In most cases, the kube-scheduler does not need to be externally accessible. Setting the bind address to `localhost` is a secure practice.
- `permit-address-sharing`: Set this to `false` to disable connection sharing through `SO_REUSEADDR`. `SO_REUSEADDR` can lead to reuse of terminated connections that are in `TIME_WAIT` state.
- `permit-port-sharing`: Default `false`. Use the default unless you are confident you understand the security implications.

Scheduler TLS command line options

- `tls-cipher-suites`: Always provide a list of preferred cipher suites. This ensures encryption never happens with insecure cipher suites.

Scheduling configurations for custom schedulers

When using custom schedulers based on the Kubernetes scheduling code, cluster administrators need to be careful with plugins that use the `queueSort`, `prefilter`, `filter`, or `permit` extension points. These extension points control various stages of a scheduling process, and the wrong configuration can impact the kube-scheduler's behavior in your cluster.

Key considerations

- Exactly one plugin that uses the `queueSort` extension point can be enabled at a time. Any plugins that use `queueSort` should be scrutinized.
- Plugins that implement the `prefilter` or `filter` extension point can potentially mark all nodes as unschedulable. This can bring scheduling of new pods to a halt.
- Plugins that implement the `permit` extension point can prevent or delay the binding of a Pod. Such plugins should be thoroughly reviewed by the cluster administrator.

When using a plugin that is not one of the default plugins, consider disabling the `queueSort`, `filter` and `permit` extension points as follows:

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
  - schedulerName: my-scheduler
    plugins:
      # Disable specific plugins for different extension points
      # You can disable all plugins for an extension point using "*"
      queueSort:
        disabled:
        - name: "*"              # Disable all queueSort plugins
      # - name: "PrioritySort"  # Disable specific queueSort plugin
      filter:
        disabled:
        - name: "*"                  # Disable all filter plugins
      # - name: "NodeResourcesFit"  # Disable specific filter plugin
      permit:
        disabled:
        - name: "*"                # Disables all permit plugins
      # - name: "TaintToleration" # Disable specific permit plugin
```

This creates a scheduler profile `my-scheduler`. Whenever the `.spec` of a Pod does not have a value for `.spec.schedulerName`, the kube-scheduler runs for that Pod, using its main configuration, and default plugins. If you define a Pod with `.spec.schedulerName` set to `my-scheduler`, the kube-scheduler runs but with a custom configuration; in that custom configuration, the `queueSort`, `filter` and `permit` extension points are disabled. If you use this KubeSchedulerConfiguration, and don't run any custom scheduler, and you then define a Pod with `.spec.schedulerName` set to `nonexistent-scheduler` (or any other scheduler name that doesn't exist in your cluster), no events would be generated for a pod.

Disallow labeling nodes

A cluster administrator should ensure that cluster users cannot label the nodes. A malicious actor can use `nodeSelector` to schedule workloads on nodes where those workloads should not be present.

15 - Kubernetes API Server Bypass Risks

Security architecture information relating to the API server and other components

The Kubernetes API server is the main point of entry to a cluster for external parties (users and services) interacting with it.

As part of this role, the API server has several key built-in security controls, such as audit logging and admission controllers. However, there are ways to modify the configuration or content of the cluster that bypass these controls.

This page describes the ways in which the security controls built into the Kubernetes API server can be bypassed, so that cluster operators and security architects can ensure that these bypasses are appropriately restricted.

Static Pods

The kubelet on each node loads and directly manages any manifests that are stored in a named directory or fetched from a specific URL as *static Pods* in your cluster. The API server doesn't manage these static Pods. An attacker with write access to this location could modify the configuration of static pods loaded from that source, or could introduce new static Pods.

Static Pods are restricted from accessing other objects in the Kubernetes API. For example, you can't configure a static Pod to mount a Secret from the cluster. However, these Pods can take other security sensitive actions, such as using `hostPath` mounts from the underlying node.

By default, the kubelet creates a mirror pod so that the static Pods are visible in the Kubernetes API. However, if the attacker uses an invalid namespace name when creating the Pod, it will not be visible in the Kubernetes API and can only be discovered by tooling that has access to the affected host(s).

If a static Pod fails admission control, the kubelet won't register the Pod with the API server. However, the Pod still runs on the node. For more information, refer to kubeadm issue #1541.

Mitigations

- Only enable the kubelet static Pod manifest functionality if required by the node.
- If a node uses the static Pod functionality, restrict filesystem access to the static Pod manifest directory or URL to users who need the access.
- Restrict access to kubelet configuration parameters and files to prevent an attacker setting a static Pod path or URL.
- Regularly audit and centrally report all access to directories or web storage locations that host static Pod manifests and kubelet configuration files.

The kubelet API

The kubelet provides an HTTP API that is typically exposed on TCP port 10250 on cluster worker nodes. The API might also be exposed on control plane nodes depending on the Kubernetes distribution in use. Direct access to the API allows for disclosure of information about the pods running on a node, the logs from those pods, and execution of commands in every container running on the node.

Some of these endpoints support Websocket protocols via HTTP `GET` requests, which are authorized with the **get** verb. This means that **get** permission on `nodes/proxy` is not a read-only permission, and authorizes access to endpoints which can be used to execute commands in any container running on the node.

When Kubernetes cluster users have RBAC access to `Node` object sub-resources, that access serves as authorization to interact with the kubelet API. The exact access depends on which sub-resource access has been granted, as detailed in kubelet authorization.

Direct access to the kubelet API is not subject to admission control and is not logged by Kubernetes audit logging. An attacker with direct access to this API may be able to bypass controls that detect or prevent certain actions.

The kubelet API can be configured to authenticate requests in a number of ways. By default, the kubelet configuration allows anonymous access. Most Kubernetes providers change the default to use webhook and certificate authentication. This lets the control plane ensure that the caller is authorized to access the `nodes` API resource or sub-resources. The default anonymous access doesn't make this assertion with the control plane.

Mitigations

- Restrict access to sub-resources of the `nodes` API object using mechanisms such as RBAC. Only grant this access when required, such as by monitoring services.

- Avoid granting the `nodes/proxy` catch-all permission, even with just the **get** verb. Instead, grant granular permissions.
- Restrict access to the kubelet port. Only allow specified and trusted IP address ranges to access the port.
- Ensure that kubelet authentication. is set to webhook or certificate mode.
- Ensure that the unauthenticated "read-only" Kubelet port is not enabled on the cluster.

The etcd API

Kubernetes clusters use etcd as a datastore. The `etcd` service listens on TCP port 2379. The only clients that need access are the Kubernetes API server and any backup tooling that you use. Direct access to this API allows for disclosure or modification of any data held in the cluster.

Access to the etcd API is typically managed by client certificate authentication. Any certificate issued by a certificate authority that etcd trusts allows full access to the data stored inside etcd.

Direct access to etcd is not subject to Kubernetes admission control and is not logged by Kubernetes audit logging. An attacker who has read access to the API server's etcd client certificate private key (or can create a new trusted client certificate) can gain cluster admin rights by accessing cluster secrets or modifying access rules. Even without elevating their Kubernetes RBAC privileges, an attacker who can modify etcd can retrieve any API object or create new workloads inside the cluster.

Many Kubernetes providers configure etcd to use mutual TLS (both client and server verify each other's certificate for authentication). There is no widely accepted implementation of authorization for the etcd API, although the feature exists. Since there is no authorization model, any certificate with client access to etcd can be used to gain full access to etcd. Typically, etcd client certificates that are only used for health checking can also grant full read and write access.

Mitigations

- Ensure that the certificate authority trusted by etcd is used only for the purposes of authentication to that service.
- Control access to the private key for the etcd server certificate, and to the API server's client certificate and key.

- Consider restricting access to the etcd port at a network level, to only allow access from specified and trusted IP address ranges.

Container runtime socket

On each node in a Kubernetes cluster, access to interact with containers is controlled by the container runtime (or runtimes, if you have configured more than one). Typically, the container runtime exposes a Unix socket that the kubelet can access. An attacker with access to this socket can launch new containers or interact with running containers.

At the cluster level, the impact of this access depends on whether the containers that run on the compromised node have access to Secrets or other confidential data that an attacker could use to escalate privileges to other worker nodes or to control plane components.

Mitigations

- Ensure that you tightly control filesystem access to container runtime sockets. When possible, restrict this access to the `root` user.
- Isolate the kubelet from other components running on the node, using mechanisms such as Linux kernel namespaces.
- Ensure that you restrict or forbid the use of `hostPath` mounts that include the container runtime socket, either directly or by mounting a parent directory. Also `hostPath` mounts must be set as read-only to mitigate risks of attackers bypassing directory restrictions.
- Restrict user access to nodes, and especially restrict superuser access to nodes.

16 - Linux kernel security constraints for Pods and containers

Overview of Linux kernel security modules and constraints that you can use to harden your Pods and containers.

This page describes some of the security features that are built into the Linux kernel that you can use in your Kubernetes workloads. To learn how to apply these features to your Pods and containers, refer to Configure a SecurityContext for a Pod or Container. You should already be familiar with Linux and with the basics of Kubernetes workloads.

Run workloads without root privileges

When you deploy a workload in Kubernetes, use the Pod specification to restrict that workload from running as the root user on the node. You can use the Pod `securityContext` to define the specific Linux user and group for the processes in the Pod, and explicitly restrict containers from running as root users. Setting these values in the Pod manifest takes precedence over similar values in the container image, which is especially useful if you're running images that you don't own.

> **Caution:**
>
> Ensure that the user or group that you assign to the workload has the permissions required for the application to function correctly. Changing the user or group to one that doesn't have the correct permissions could lead to file access issues or failed operations.

Configuring the kernel security features on this page provides fine-grained control over the actions that processes in your cluster can take, but managing these configurations can be challenging at scale. Running containers as non-root, or in user namespaces if you need root privileges, helps to reduce the chance that you'll need to enforce your configured kernel security capabilities.

Security features in the Linux kernel

Kubernetes lets you configure and use Linux kernel features to improve isolation and harden your containerized workloads. Common features include the following:

- **Secure computing mode (seccomp)**: Filter which system calls a process can make
- **AppArmor**: Restrict the access privileges of individual programs
- **Security Enhanced Linux (SELinux)**: Assign security labels to objects for more manageable security policy enforcement

To configure settings for one of these features, the operating system that you choose for your nodes must enable the feature in the kernel. For example, Ubuntu 7.10 and later enable AppArmor by default. To learn whether your OS enables a specific feature, consult the OS documentation.

You use the `securityContext` field in your Pod specification to define the constraints that apply to those processes. The `securityContext` field also supports other security settings, such as specific Linux capabilities or file access permissions using UIDs and GIDs. To learn more, refer to Configure a SecurityContext for a Pod or Container.

seccomp

Some of your workloads might need privileges to perform specific actions as the root user on your node's host machine. Linux uses *capabilities* to divide the available privileges into categories, so that processes can get the privileges required to perform specific actions without being granted all privileges. Each capability has a set of system calls (syscalls) that a process can make. seccomp lets you restrict these individual syscalls. It can be used to sandbox the privileges of a process, restricting the calls it is able to make from userspace into the kernel.

In Kubernetes, you use a *container runtime* on each node to run your containers. Example runtimes include CRI-O, Docker, or containerd. Each runtime allows only a subset of Linux capabilities by default. You can further limit the allowed syscalls individually by using a seccomp profile. Container runtimes usually include a default seccomp profile. Kubernetes lets you automatically apply seccomp profiles loaded onto a node to your Pods and containers.

> **Note:**
> Kubernetes also has the `allowPrivilegeEscalation` setting for Pods and containers. When set to `false`, this prevents processes from gaining new capabilities and restricts unprivileged users from changing the applied seccomp profile to a more permissive profile.

To learn how to implement seccomp in Kubernetes, refer to Restrict a Container's Syscalls with seccomp or the Seccomp node reference

To learn more about seccomp, see Seccomp BPF in the Linux kernel documentation.

Considerations for seccomp

seccomp is a low-level security configuration that you should only configure yourself if you require fine-grained control over Linux syscalls. Using seccomp, especially at scale, has the following risks:

- Configurations might break during application updates
- Attackers can still use allowed syscalls to exploit vulnerabilities
- Profile management for individual applications becomes challenging at scale

Recommendation: Use the default seccomp profile that's bundled with your container runtime. If you need a more isolated environment, consider using a sandbox, such as gVisor. Sandboxes solve the preceding risks with custom seccomp profiles, but require more compute resources on your nodes and might have compatibility issues with GPUs and other specialized hardware.

AppArmor and SELinux: policy-based mandatory access control

You can use Linux policy-based mandatory access control (MAC) mechanisms, such as AppArmor and SELinux, to harden your Kubernetes workloads.

AppArmor

AppArmor is a Linux kernel security module that supplements the standard Linux user and group based permissions to confine programs to a limited set of resources. AppArmor can be configured for any application to reduce its potential attack surface and provide greater in-depth defense. It is configured through profiles tuned to allow the access needed by a specific program or container, such as Linux capabilities, network access, and file permissions. Each profile can be run in either enforcing mode, which blocks access to disallowed resources, or complain mode, which only reports violations.

AppArmor can help you to run a more secure deployment by restricting what containers are allowed to do, and/or provide better auditing through system logs. The container runtime that you use might ship with a default AppArmor profile, or you can use a custom profile.

To learn how to use AppArmor in Kubernetes, refer to Restrict a Container's Access to Resources with AppArmor.

SELinux

SELinux is a Linux kernel security module that lets you restrict the access that a specific *subject*, such as a process, has to the files on your system. You define security policies that apply to subjects that have specific SELinux labels. When a process that has an SELinux label attempts to access a file, the SELinux server checks whether that process' security policy allows the access and makes an authorization decision.

In Kubernetes, you can set an SELinux label in the `securityContext` field of your manifest. The specified labels are assigned to those processes. If you have configured security policies that affect those labels, the host OS kernel enforces these policies.

To learn how to use SELinux in Kubernetes, refer to Assign SELinux labels to a container.

Differences between AppArmor and SELinux

The operating system on your Linux nodes usually includes one of either AppArmor or SELinux. Both mechanisms provide similar types of protection, but have differences such as the following:

- **Configuration**: AppArmor uses profiles to define access to resources. SELinux uses policies that apply to specific labels.
- **Policy application**: In AppArmor, you define resources using file paths. SELinux uses the index node (inode) of a resource to identify the resource.

Summary of features

The following table describes the use cases and scope of each security control. You can use all of these controls together to build a more hardened system.

Security feature	Description	How to use	Example
seccomp	Restrict individual kernel calls in the userspace. Reduces the likelihood that a vulnerability that uses a	Specify a loaded seccomp profile in the Pod or container specification to apply its constraints to the processes in the Pod.	Reject the `unshare` syscall, which was used in CVE-2022-0185.

Security feature	Description	How to use	Example
	restricted syscall would compromise the system.		
AppArmor	Restrict program access to specific resources. Reduces the attack surface of the program. Improves audit logging.	Specify a loaded AppArmor profile in the container specification.	Restrict a read-only program from writing to any file path in the system.
SELinux	Restrict access to resources such as files, applications, ports, and processes using labels and security policies.	Specify access restrictions for specific labels. Tag processes with those labels to enforce the access restrictions related to the label.	Restrict a container from accessing files outside its own filesystem.

Summary of Linux kernel security features

> **Note:**
> Mechanisms like AppArmor and SELinux can provide protection that extends beyond the container. For example, you can use SELinux to help mitigate CVE-2019-5736.

Considerations for managing custom configurations

seccomp, AppArmor, and SELinux usually have a default configuration that offers basic protections. You can also create custom profiles and policies that meet the requirements of your workloads. Managing and distributing these custom configurations at scale might be challenging, especially if you use all three features together. To help you to manage these configurations at scale, use a tool like the Kubernetes Security Profiles Operator.

Kernel-level security features and privileged containers

Kubernetes lets you specify that some trusted containers can run in *privileged* mode. Any container in a Pod can run in privileged mode to use operating system administrative capabilities that would otherwise be inaccessible. This is available for both Windows and Linux.

Privileged containers explicitly override some of the Linux kernel constraints that you might use in your workloads, as follows:

- **seccomp**: Privileged containers run as the `Unconfined` seccomp profile, overriding any seccomp profile that you specified in your manifest.
- **AppArmor**: Privileged containers ignore any applied AppArmor profiles.
- **SELinux**: Privileged containers run as the `unconfined_t` domain.

Privileged containers

Any container in a Pod can enable *Privileged mode* if you set the `privileged: true` field in the `securityContext` field for the container. Privileged containers override or undo many other hardening settings such as the applied seccomp profile, AppArmor profile, or SELinux constraints. Privileged containers are given all Linux capabilities, including capabilities that they don't require. For example, a root user in a privileged container might be able to use the `CAP_SYS_ADMIN` and `CAP_NET_ADMIN` capabilities on the node, bypassing the runtime seccomp configuration and other restrictions.

In most cases, you should avoid using privileged containers, and instead grant the specific capabilities required by your container using the `capabilities` field in the `securityContext` field. Only use privileged mode if you have a capability that you can't grant with the securityContext. This is useful for containers that want to use operating system administrative capabilities such as manipulating the network stack or accessing hardware devices.

In Kubernetes version 1.26 and later, you can also run Windows containers in a similarly privileged mode by setting the `windowsOptions.hostProcess` flag on the security context of the Pod spec. For details and instructions, see Create a Windows HostProcess Pod.

Recommendations and best practices

- Before configuring kernel-level security capabilities, you should consider implementing network-level isolation. For more information, read the Security Checklist.

- Unless necessary, run Linux workloads as non-root by setting specific user and group IDs in your Pod manifest and by specifying `runAsNonRoot: true`.

Additionally, you can run workloads in user namespaces by setting `hostUsers: false` in your Pod manifest. This lets you run containers as root users in the user namespace, but as non-root users in the host namespace on the node. This is still in early stages of development and might not have the level of support that you need. For instructions, refer to Use a User Namespace With a Pod.

What's next

- Learn how to use AppArmor
- Learn how to use seccomp
- Learn how to use SELinux
- Seccomp Node Reference

17 - Security Checklist

Baseline checklist for ensuring security in Kubernetes clusters.

This checklist aims at providing a basic list of guidance with links to more comprehensive documentation on each topic. It does not claim to be exhaustive and is meant to evolve.

On how to read and use this document:

- The order of topics does not reflect an order of priority.
- Some checklist items are detailed in the paragraph below the list of each section.

> **Caution:**
>
> Checklists are **not** sufficient for attaining a good security posture on their own. A good security posture requires constant attention and improvement, but a checklist can be the first step on the never-ending journey towards security preparedness. Some of the recommendations in this checklist may be too restrictive or too lax for your specific security needs. Since Kubernetes security is not "one size fits all", each category of checklist items should be evaluated on its merits.

Authentication & Authorization

- ☐ `system:masters` group is not used for user or component authentication after bootstrapping.
- ☐ The kube-controller-manager is running with `--use-service-account-credentials` enabled.
- ☐ The root certificate is protected (either an offline CA, or a managed online CA with effective access controls).
- ☐ Intermediate and leaf certificates have an expiry date no more than 3 years in the future.
- ☐ A process exists for periodic access review, and reviews occur no more than 24 months apart.
- ☐ The Role Based Access Control Good Practices are followed for guidance related to authentication and authorization.

After bootstrapping, neither users nor components should authenticate to the Kubernetes API as `system:masters`. Similarly, running all of kube-controller-manager as

`system:masters` should be avoided. In fact, `system:masters` should only be used as a break-glass mechanism, as opposed to an admin user.

Network security

- ☐ CNI plugins in use support network policies.
- ☐ Ingress and egress network policies are applied to all workloads in the cluster.
- ☐ Default network policies within each namespace, selecting all pods, denying everything, are in place.
- ☐ If appropriate, a service mesh is used to encrypt all communications inside of the cluster.
- ☐ The Kubernetes API, kubelet API and etcd are not exposed publicly on Internet.
- ☐ Access from the workloads to the cloud metadata API is filtered.
- ☐ Use of LoadBalancer and ExternalIPs is restricted.

A number of Container Network Interface (CNI) plugins plugins provide the functionality to restrict network resources that pods may communicate with. This is most commonly done through Network Policies which provide a namespaced resource to define rules. Default network policies that block all egress and ingress, in each namespace, selecting all pods, can be useful to adopt an allow list approach to ensure that no workloads are missed.

Not all CNI plugins provide encryption in transit. If the chosen plugin lacks this feature, an alternative solution could be to use a service mesh to provide that functionality.

The etcd datastore of the control plane should have controls to limit access and not be publicly exposed on the Internet. Furthermore, mutual TLS (mTLS) should be used to communicate securely with it. The certificate authority for this should be unique to etcd.

External Internet access to the Kubernetes API server should be restricted to not expose the API publicly. Be careful, as many managed Kubernetes distributions are publicly exposing the API server by default. You can then use a bastion host to access the server.

The kubelet API access should be restricted and not exposed publicly, the default authentication and authorization settings, when no configuration file specified with the `--config` flag, are overly permissive.

If a cloud provider is used for hosting Kubernetes, the access from pods to the cloud metadata API `169.254.169.254` should also be restricted or blocked if not needed because it may leak information.

For restricted LoadBalancer and ExternalIPs use, see CVE-2020-8554: Man in the middle using LoadBalancer or ExternalIPs and the DenyServiceExternalIPs admission controller for further information.

Pod security

- ☐ RBAC rights to `create`, `update`, `patch`, `delete` workloads is only granted if necessary.
- ☐ Appropriate Pod Security Standards policy is applied for all namespaces and enforced.
- ☐ Memory limit is set for the workloads with a limit equal or inferior to the request.
- ☐ CPU limit might be set on sensitive workloads.
- ☐ For nodes that support it, Seccomp is enabled with appropriate syscalls profile for programs.
- ☐ For nodes that support it, AppArmor or SELinux is enabled with appropriate profile for programs.

RBAC authorization is crucial but cannot be granular enough to have authorization on the Pods' resources (or on any resource that manages Pods). The only granularity is the API verbs on the resource itself, for example, `create` on Pods. Without additional admission, the authorization to create these resources allows direct unrestricted access to the schedulable nodes of a cluster.

The Pod Security Standards define three different policies, privileged, baseline and restricted that limit how fields can be set in the `PodSpec` regarding security. These standards can be enforced at the namespace level with the new Pod Security admission, enabled by default, or by third-party admission webhook. Please note that, contrary to the removed PodSecurityPolicy admission it replaces, Pod Security admission can be easily combined with admission webhooks and external services.

Pod Security admission `restricted` policy, the most restrictive policy of the Pod Security Standards set, can operate in several modes, `warn`, `audit` or `enforce` to gradually apply the most appropriate security context according to security best practices. Nevertheless, pods' security context should be separately investigated to limit the privileges and access pods may have on top of the predefined security standards, for specific use cases.

For a hands-on tutorial on Pod Security, see the blog post Kubernetes 1.23: Pod Security Graduates to Beta.

Memory and CPU limits should be set in order to restrict the memory and CPU resources a pod can consume on a node, and therefore prevent potential DoS attacks from malicious or breached workloads. Such policy can be enforced by an admission controller. Please note that CPU limits will throttle usage and thus can have unintended effects on auto-scaling features or efficiency i.e. running the process in best effort with the CPU resource available.

> **Caution:**
> Memory limit superior to request can expose the whole node to OOM issues.

Enabling Seccomp

Seccomp stands for secure computing mode and has been a feature of the Linux kernel since version 2.6.12. It can be used to sandbox the privileges of a process, restricting the calls it is able to make from userspace into the kernel. Kubernetes lets you automatically apply seccomp profiles loaded onto a node to your Pods and containers.

Seccomp can improve the security of your workloads by reducing the Linux kernel syscall attack surface available inside containers. The seccomp filter mode leverages BPF to create an allow or deny list of specific syscalls, named profiles.

Since Kubernetes 1.27, you can enable the use of `RuntimeDefault` as the default seccomp profile for all workloads. A security tutorial is available on this topic. In addition, the Kubernetes Security Profiles Operator is a project that facilitates the management and use of seccomp in clusters.

> **Note:**
> Seccomp is only available on Linux nodes.

Enabling AppArmor or SELinux

AppArmor

AppArmor is a Linux kernel security module that can provide an easy way to implement Mandatory Access Control (MAC) and better auditing through system logs. A default AppArmor profile is enforced on nodes that support it, or a custom profile can be configured. Like seccomp, AppArmor is also configured through profiles, where each

profile is either running in enforcing mode, which blocks access to disallowed resources or complain mode, which only reports violations. AppArmor profiles are enforced on a per-container basis, with an annotation, allowing for processes to gain just the right privileges.

> **Note:**
> AppArmor is only available on Linux nodes, and enabled in some Linux distributions.

SELinux

SELinux is also a Linux kernel security module that can provide a mechanism for supporting access control security policies, including Mandatory Access Controls (MAC). SELinux labels can be assigned to containers or pods via their `securityContext` section.

> **Note:**
> SELinux is only available on Linux nodes, and enabled in some Linux distributions.

Logs and auditing

- ☐ Audit logs, if enabled, are protected from general access.

Pod placement

- ☐ Pod placement is done in accordance with the tiers of sensitivity of the application.
- ☐ Sensitive applications are running isolated on nodes or with specific sandboxed runtimes.

Pods that are on different tiers of sensitivity, for example, an application pod and the Kubernetes API server, should be deployed onto separate nodes. The purpose of node isolation is to prevent an application container breakout to directly providing access to applications with higher level of sensitivity to easily pivot within the cluster. This separation should be enforced to prevent pods accidentally being deployed onto the same node. This could be enforced with the following features:

Node Selectors

Key-value pairs, as part of the pod specification, that specify which nodes to deploy onto. These can be enforced at the namespace and cluster level with the PodNodeSelector admission controller.

PodTolerationRestriction

An admission controller that allows administrators to restrict permitted tolerations within a namespace. Pods within a namespace may only utilize the tolerations specified on the namespace object annotation keys that provide a set of default and allowed tolerations.

RuntimeClass

RuntimeClass is a feature for selecting the container runtime configuration. The container runtime configuration is used to run a Pod's containers and can provide more or less isolation from the host at the cost of performance overhead.

Secrets

- ☐ ConfigMaps are not used to hold confidential data.
- ☐ Encryption at rest is configured for the Secret API.
- ☐ If appropriate, a mechanism to inject secrets stored in third-party storage is deployed and available.
- ☐ Service account tokens are not mounted in pods that don't require them.
- ☐ Bound service account token volume is in-use instead of non-expiring tokens.

Secrets required for pods should be stored within Kubernetes Secrets as opposed to alternatives such as ConfigMap. Secret resources stored within etcd should be encrypted at rest.

Pods needing secrets should have these automatically mounted through volumes, preferably stored in memory like with the `emptyDir.medium` option. Mechanism can be used to also inject secrets from third-party storages as volume, like the Secrets Store CSI Driver. This should be done preferentially as compared to providing the pods service account RBAC access to secrets. This would allow adding secrets into the pod as environment variables or files. Please note that the environment variable method might be more prone to leakage due to crash dumps in logs and the non-confidential nature of environment variable in Linux, as opposed to the permission mechanism on files.

Service account tokens should not be mounted into pods that do not require them. This can be configured by setting `automountServiceAccountToken` to `false` either within the

service account to apply throughout the namespace or specifically for a pod. For Kubernetes v1.22 and above, use Bound Service Accounts for time-bound service account credentials.

Images

- ☐ Minimize unnecessary content in container images.
- ☐ Container images are configured to be run as unprivileged user.
- ☐ References to container images are made by sha256 digests (rather than tags) or the provenance of the image is validated by verifying the image's digital signature at deploy time via admission control.
- ☐ Container images are regularly scanned during creation and in deployment, and known vulnerable software is patched.

Container image should contain the bare minimum to run the program they package. Preferably, only the program and its dependencies, building the image from the minimal possible base. In particular, image used in production should not contain shells or debugging utilities, as an ephemeral debug container can be used for troubleshooting.

Build images to directly start with an unprivileged user by using the `USER` instruction in Dockerfile. The Security Context allows a container image to be started with a specific user and group with `runAsUser` and `runAsGroup`, even if not specified in the image manifest. However, the file permissions in the image layers might make it impossible to just start the process with a new unprivileged user without image modification.

Avoid using image tags to reference an image, especially the `latest` tag, the image behind a tag can be easily modified in a registry. Prefer using the complete `sha256` digest which is unique to the image manifest. This policy can be enforced via an ImagePolicyWebhook. Image signatures can also be automatically verified with an admission controller at deploy time to validate their authenticity and integrity.

Scanning a container image can prevent critical vulnerabilities from being deployed to the cluster alongside the container image. Image scanning should be completed before deploying a container image to a cluster and is usually done as part of the deployment process in a CI/CD pipeline. The purpose of an image scan is to obtain information about possible vulnerabilities and their prevention in the container image, such as a Common Vulnerability Scoring System (CVSS) score. If the result of the image scans is combined with the pipeline compliance rules, only properly patched container images will end up in Production.

Admission controllers

- ☐ An appropriate selection of admission controllers is enabled.
- ☐ A pod security policy is enforced by the Pod Security Admission or/and a webhook admission controller.
- ☐ The admission chain plugins and webhooks are securely configured.

Admission controllers can help improve the security of the cluster. However, they can present risks themselves as they extend the API server and should be properly secured.

The following lists present a number of admission controllers that could be considered to enhance the security posture of your cluster and application. It includes controllers that may be referenced in other parts of this document.

This first group of admission controllers includes plugins enabled by default, consider to leave them enabled unless you know what you are doing:

`CertificateApproval`

Performs additional authorization checks to ensure the approving user has permission to approve certificate request.

`CertificateSigning`

Performs additional authorization checks to ensure the signing user has permission to sign certificate requests.

`CertificateSubjectRestriction`

Rejects any certificate request that specifies a 'group' (or 'organization attribute') of `system:masters`.

`LimitRanger`

Enforces the LimitRange API constraints.

`MutatingAdmissionWebhook`

Allows the use of custom controllers through webhooks, these controllers may mutate requests that they review.

`PodSecurity`

Replacement for Pod Security Policy, restricts security contexts of deployed Pods.

`ResourceQuota`

Enforces resource quotas to prevent over-usage of resources.

`ValidatingAdmissionWebhook`

Allows the use of custom controllers through webhooks, these controllers do not mutate requests that it reviews.

The second group includes plugins that are not enabled by default but are in general availability state and are recommended to improve your security posture:

`DenyServiceExternalIPs`

Rejects all net-new usage of the `Service.spec.externalIPs` field. This is a mitigation for CVE-2020-8554: Man in the middle using LoadBalancer or ExternalIPs.

`NodeRestriction`

Restricts kubelet's permissions to only modify the pods API resources they own or the node API resource that represent themselves. It also prevents kubelet from using the `node-restriction.kubernetes.io/` annotation, which can be used by an attacker with access to the kubelet's credentials to influence pod placement to the controlled node.

The third group includes plugins that are not enabled by default but could be considered for certain use cases:

`AlwaysPullImages`

Enforces the usage of the latest version of a tagged image and ensures that the deployer has permissions to use the image.

`ImagePolicyWebhook`

Allows enforcing additional controls for images through webhooks.

What's next

- Privilege escalation via Pod creation warns you about a specific access control risk; check how you're managing that threat.
  - If you use Kubernetes RBAC, read RBAC Good Practices for further information on authorization.
- Securing a Cluster for information on protecting a cluster from accidental or malicious access.

- Cluster Multi-tenancy guide for configuration options recommendations and best practices on multi-tenancy.
- Blog post "A Closer Look at NSA/CISA Kubernetes Hardening Guidance" for complementary resource on hardening Kubernetes clusters.

18 - Application Security Checklist

Baseline guidelines around ensuring application security on Kubernetes, aimed at application developers

This checklist aims to provide basic guidelines on securing applications running in Kubernetes from a developer's perspective. This list is not meant to be exhaustive and is intended to evolve over time.

On how to read and use this document:

- The order of topics does not reflect an order of priority.
- Some checklist items are detailed in the paragraph below the list of each section.
- This checklist assumes that a `developer` is a Kubernetes cluster user who interacts with namespaced scope objects.

> **Caution:**
>
> Checklists are **not** sufficient for attaining a good security posture on their own. A good security posture requires constant attention and improvement, but a checklist can be the first step on the never-ending journey towards security preparedness. Some recommendations in this checklist may be too restrictive or too lax for your specific security needs. Since Kubernetes security is not "one size fits all", each category of checklist items should be evaluated on its merits.

Base security hardening

The following checklist provides base security hardening recommendations that would apply to most applications deploying to Kubernetes.

Application design

- ☐ Follow the right security principles when designing applications.
- ☐ Application configured with appropriate QoS class through resource request and limits.
  - ☐ Memory limit is set for the workloads with a limit equal to or greater than the request.
  - ☐ CPU limit might be set on sensitive workloads.

Service account

- ☐ Avoid using the `default` ServiceAccount. Instead, create ServiceAccounts for each workload or microservice.
- ☐ `automountServiceAccountToken` should be set to `false` unless the pod specifically requires access to the Kubernetes API to operate.

Pod-level `securityContext` recommendations

- ☐ Set `runAsNonRoot: true`.
- ☐ Configure the container to execute as a less privileged user (for example, using `runAsUser` and `runAsGroup`), and configure appropriate permissions on files or directories inside the container image.
- ☐ Optionally add a supplementary group with `fsGroup` to access persistent volumes.
- ☐ The application deploys into a namespace that enforces an appropriate Pod security standard. If you cannot control this enforcement for the cluster(s) where the application is deployed, take this into account either through documentation or additional defense in depth.

Container-level `securityContext` recommendations

- ☐ Disable privilege escalations using `allowPrivilegeEscalation: false`.
- ☐ Configure the root filesystem to be read-only with `readOnlyRootFilesystem: true`.
- ☐ Avoid running privileged containers (set `privileged: false`).
- ☐ Drop all capabilities from the containers and add back only specific ones that are needed for operation of the container.

Role Based Access Control (RBAC)

- ☐ Permissions such as **create**, **patch**, **update** and **delete** should be only granted if necessary.
- ☐ Avoid creating RBAC permissions to create or update roles which can lead to privilege escalation.
- ☐ Review bindings for the `system:unauthenticated` group and remove them where possible, as this gives access to anyone who can contact the API server at a network level.

The **create**, **update** and **delete** verbs should be permitted judiciously. The **patch** verb if allowed on a Namespace can allow users to update labels on the namespace or

deployments which can increase the attack surface.

For sensitive workloads, consider providing a recommended ValidatingAdmissionPolicy that further restricts the permitted write actions.

Image security

- ☐ Using an image scanning tool to scan an image before deploying containers in the Kubernetes cluster.
- ☐ Use container signing to validate the container image signature before deploying to the Kubernetes cluster.

Network policies

- ☐ Configure NetworkPolicies to only allow expected ingress and egress traffic from the pods.

Make sure that your cluster provides and enforces NetworkPolicy. If you are writing an application that users will deploy to different clusters, consider whether you can assume that NetworkPolicy is available and enforced.

Advanced security hardening

This section of this guide covers some advanced security hardening points which might be valuable based on different Kubernetes environment setup.

Linux container security

Configure Security Context for the pod-container.

- ☐ Set the Seccomp Profile for a Container.
- ☐ Restrict a Container's Access to Resources with AppArmor.
- ☐ Assign SELinux Labels to a Container.

Runtime classes

- ☐ Configure appropriate runtime classes for containers.

> **Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the content guide before submitting a change. More information.

Some containers may require a different isolation level from what is provided by the default runtime of the cluster. `runtimeClassName` can be used in a podspec to define a different runtime class.

For sensitive workloads consider using kernel emulation tools like gVisor, or virtualized isolation using a mechanism such as kata-containers.

In high trust environments, consider using confidential virtual machines to improve cluster security even further.